Queen's University

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Pre-Publication Ambiguity Detection in Internet-Drafts

Revision-Aware Machine Learning and Large Language Models

Master's Project Report

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Abstract

Natural-language protocol specifications can permit multiple plausible implementations even when they use the normative keywords of BCP 14. Detecting such ambiguity before an Internet-Draft becomes an RFC is difficult because clauses recur across revisions, positive cases are rare, and final RFC or erratum evidence can leak into draft-stage prediction. This report presents DraftAmbiguity, a revision-aware corpus, annotation workflow, modeling pipeline, and local review prototype designed around temporal information boundaries and draft-family-isolated evaluation. The corpus contains 253 draft revisions, 12 RFCs, and 31 verified technical errata. Separate development studies evaluate a deterministic heuristic, classical machine learning, frozen embeddings, contemporaneous revision features, Qwen3 prompting and lexical–semantic hybrids, and DeBERTa variants. On the latest 2,425-row matched classical cohort (160 positives), engineered features plus TF–IDF obtained AP 0.3584 with a Linear SVM, 0.3363 with a random forest, and 0.3265 with logistic regression. Adding twelve past-only revision features changed these values to 0.3572, 0.3348, and 0.3185, respectively, so revision history did not improve ranking AP in this rerun. On the separate 2,143-clause reviewed-label transformer development cohort, a fitting-side ECDF ensemble of two DeBERTa variants obtained AP 0.6736, compared with 0.5823 and 0.5761 for its components. These confidence intervals overlap, only one seed was run, and 100 human-confirmed records were selected from historical errors; the result is descriptive rather than an independent generalization estimate. A separate Qwen/MiniLM hybrid reached AP 0.3252, but the intended conditional LLM-routing, standardized latency/cost comparison, and expert evaluation of explanation faithfulness were not completed. Moreover, the latest classical rerun includes the TCP and DNS families that the proposal intended to reserve, so they can no longer serve as untouched project-level holdouts. The project therefore answers the proposal's effectiveness and revision-ablation questions with qualified development evidence, while the LLM-call-reduction and explanation-quality questions remain open. DraftAmbiguity should be treated as a human-review prioritization aid, not an autonomous ambiguity judge.

Keywords: Internet-Drafts, RFC, ambiguity detection, protocol specifications, natural language processing, DeBERTa, document evolution, leakage control.

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Chapter 1

Introduction

1.1 Background and Problem Statement

The Internet Engineering Task Force (IETF) develops protocols through sequences of mutable Internet-Drafts before stable RFC publication. Although capitalized terms such as *MUST*, *SHOULD*, and *MAY* have defined meanings under BCP 14 [1, 2], their surrounding prose can still leave actors, conditions, references, failure behavior, or interactions under-specified. Multiple standards-conforming interpretations can then produce interoperability defects or security-relevant implementation differences.

Prior systems show that protocol prose can be analyzed automatically. SAGE combines semantic parsing and author clarification to disambiguate RFC text and generate code [3]. NRFCKG represents protocol requirements as a knowledge graph and searches for contradictions [4]. Most directly, RFCScope derives seven logical-ambiguity subtypes from verified errata and uses an LLM pipeline to find ambiguities in published RFCs [5]. These systems motivate automated support, but they do not answer a different operational question: *could a risky clause have been prioritized while it was still an Internet-Draft, using only information available at that time?*

1.2 Motivation

That question introduces three hazards. First, a final RFC, later revision, or later erratum is an informative label source but an invalid prediction-time feature. Second, clause-level random splits place nearly identical text from the same revision lineage on both sides of an evaluation. Third, the positive class is rare, so accuracy and even ROC area can obscure poor retrieval of ambiguous clauses; precision–recall analysis is more informative under skew [6, 7].

1.3 Aim, Objectives, and Research Questions

This paper reports a complete development-stage study of DraftAmbiguity. Its central design principle is separation: source collection, structural parsing, revision alignment, candidate construction, annotation, model input, model selection, reserved validation, and sealed testing are distinct, hash-recorded stages. The intended output is not an automatic declaration that a standard is defective. It is a ranking that helps an author or reviewer allocate attention.

The project evaluates the five research questions defined in the approved proposal:

RQ1: How effectively can ambiguous requirements be detected in historical Internet-Drafts using only information available at the evaluated draft version?

RQ2: Do contemporaneous revision-history and cross-reference features improve F_2, PR-AUC, and calibration over text-only models?

RQ3: How do heuristic, classical ML, fine-tuned transformer, LLM-only, and hybrid approaches compare in effectiveness, reproducibility, latency, cost, and review workload?

RQ4: Can hybrid routing preserve or improve detection effectiveness while reducing LLM calls relative to an LLM-only approach?

RQ5: How faithful and useful are the evidence spans, ambiguity subtypes, and explanations produced by the LLM verifier?

The proposal also stated four hypotheses: revision-history features would improve F_2 and PR-AUC; hybrid routing would retain effectiveness while reducing LLM use; evidence grounding would reduce unsupported explanations; and random clause splitting would be more optimistic than family/temporal splits. This report distinguishes tested, partly tested, and untested claims rather than treating implementation as confirmation.

1.4 Scope and Contributions

The contributions are:

- A revision-aware corpus covering 253 draft revisions, 12 RFCs, and 31 verified technical errata, with future-information guards.
- A reproducible annotation workflow spanning eleven draft families, with explicit consensus rules, source provenance, and temporal-information controls.
- Matched-cohort evaluation of classical classifiers and a deterministic heuristic, plus

separate leave-one-family-out comparisons of lexical, engineered, embedding, contextual, and revision-history features.

- Separate, auditable studies of local Qwen prompting and lexical/semantic hybrids, transformer weighting and context/soft-label ablations, and a hash-pinned target-only DeBERTa-v3-base comparison.
- A project-wide accounting of early low-prevalence pilots, nested calibration and routing experiments, cross-model rank ensembles, protocol corrections, and blocked or incomplete study lines, so null and unavailable evidence is not silently omitted.
- Family-isolated OOF, nested threshold, family-robustness, and fixed-review-budget analyses, reported without conflating their distinct data artifacts.
- A claim boundary: AI-assisted labels are not adjudicated expert gold, model-vote agreement is limited, the final transformer uses one seed, and the proposal's TCP/DNS holdouts were consumed by the latest all-family classical rerun even though they remain excluded from the transformer and Qwen development studies.

The scope is binary clause-level prioritization for human review. It excludes autonomous standards certification, automatic protocol implementation, and claims that the absence of an erratum proves clarity. The completed experiments are development studies over several explicitly separated annotation snapshots. Although the transformer and Qwen studies preserve their nine-family development boundary, the latest classical rerun evaluates all eleven labeled families, including the proposed TCP and DNS holdouts. The repository therefore has no longer an untouched project-level TCP/DNS test, and no independently sampled expert-gold test set was completed.

1.5 Report Organization

Chapter 2 reviews ambiguity analysis, protocol NLP, document evolution, and evaluation under class imbalance. Chapter 3 describes the corpus, annotation, leakage controls, models, and experimental protocol. Chapter 4 presents the implemented system architecture and review workflow. Chapter 5 reports results without merging incompatible cohorts. Chapter 6 interprets the evidence and answers each proposal research question. Chapter 7 concludes and identifies the work required for a confirmatory evaluation. The appendices provide research-question traceability and reproduction instructions.

Chapter 2

Background and Literature Review

2.1 Ambiguity in Requirements and Protocol Text

Ambiguity detection in requirements engineering includes lexical/rule-based indicators, supervised classifiers, and contextual language models. TAPHSIR combines machine learning for anaphoric-ambiguity detection with a BERT-based resolution component [8]; a recent hybrid requirements study combines ambiguous-phrase rules with conventional classifiers [9]. These systems demonstrate complementary roles for interpretable cues and learned representations. Their metrics are not transferable to Internet-Drafts, however, because ambiguity definitions, annotation units, prevalence, and document-family dependencies differ. In protocol text, a phrase can be lexically vague yet harmless under a surrounding state machine, while a precise-looking clause may still omit an actor, boundary, or conflict-resolution rule.

Protocol specifications add domain structure: packet fields, state transitions, normative strength, cross-references, and compatibility requirements. SAGE converts selected RFC prose to logical forms, asks humans to resolve remaining alternatives, and generates protocol code [3]. PROSPER uses LLMs to extract protocol state machines from RFC text and diagrams [10]. NRFCKG targets direct and conditional contradictions using extracted entities, rules, and constraint solving [4]. These systems address important neighboring problems. Formalization and contradiction checking can establish specific logical properties, but they do not by themselves prioritize ordinary draft clauses for human ambiguity review.

RFCScope is the closest conceptual predecessor. It analyzes 273 verified technical errata, defines seven subtypes under inconsistency and under-specification, and uses context construction, semantic partitioning, subtype-aware LLM analysis, and validation. Its evaluation reports 31 new ambiguities in 14 of 20 RFCs, including author-confirmed cases [5]. We use its subtype vocabulary as conceptual guidance, not as an automatic label oracle. Our prediction-time unit is a clause in a historical draft revision; our bi-

nary operational question is whether an ordinary implementer could reasonably derive materially different implementation behavior from the available text.

2.2 Document Evolution and Leakage-Aware Evaluation

Scientific-writing resources such as arXivEdits and CASIMIR model document change through aligned versions and edit categories [11, 12]. Protocol specifications share the need to represent insertions, deletions, moves, splits, and merges, but a change is not itself evidence of ambiguity: revisions also contain editorial cleanup, new functionality, and terminology updates. This distinction motivates measuring alignment coverage separately from label quality and detector performance.

Evaluation must also respect the intended use. If the goal is transfer to a protocol family not seen during fitting, random clause-level splits are optimistic because clauses, templates, and lineages recur. Family-grouped evaluation is therefore more relevant than random splits, although it estimates transfer only over observed families and cannot substitute for a prospective test. Under rare-positive labels, average precision and precision–recall curves are more informative than accuracy alone [6, 7]. Our contribution relative to prior work is not a new general ambiguity taxonomy; it is an end-to-end draft-stage study that makes temporal availability, family isolation, candidate selection, source-label artifacts, and evaluation cohorts explicit.

2.3 Revision Alignment

Document evolution has been studied outside networking. arXivEdits provides multi-version scientific papers with sentence alignments and edit intentions [11]; CASIMIR extends revision corpora at larger scale [12]. Those datasets establish that insertion, deletion, movement, split, and merge must be represented explicitly. In a protocol specification, however, a textual change is not itself evidence of ambiguity: it may introduce functionality, update terminology, or merely edit style. We therefore separate *correspondence* from *ambiguity*, and never use the final RFC or future revision as a detector input.

2.4 Models for Imbalanced Text Classification

The reviewed-label experiment compares two target-only `microsoft/deberta-v3-base` classifiers and an equal-weight rank ensemble on one frozen cohort. DeBERTa-v3 uses replaced-token detection and gradient-disentangled embedding sharing [13]. Focal loss

emphasizes hard examples under class imbalance [14]. The partial model freezes embeddings, the shared backbone, and layers 0–7, and trains layers 8–11, the pooler, and classifier. Earlier local-Qwen/hybrid and classical studies are reported separately because they were evaluated under different data and validation protocols.

Table 2.1: Positioning relative to representative protocol and document-analysis work. The projects solve related but non-identical tasks.

Work	Primary task	Evidence and method	Relation to this study
SAGE [3]	Disambiguate protocol prose to generate implementations	Logical representations, author clarification, code generation	Resolves selected requirements; this work ranks draft clauses for review without generating protocol code
NRFCKG [4]	Detect protocol contradictions	Neural knowledge-graph construction and constraint-oriented analysis	Targets contradiction structure; this work studies broader clause-level ambiguity against an operational annotation reference
PROSPER [10]	Extract protocol state machines	LLM extraction from RFC prose and diagrams	Provides structured protocol understanding; it does not evaluate pre-publication clause prioritization here
RFCScope [5]	Find logical ambiguities in published RFCs	Verified-errata taxonomy, context-aware LLM analysis, and validation	Closest task predecessor; our temporal target is the historical draft-stage and our emphasis is on leakage-controlled family-held-out evidence
arXivEdits / CASIMIR [11, 12]	Model scientific-document revisions	Multi-version text and edit alignments	Motivate explicit lineage and change handling; protocol edits are not treated as ambiguity labels
DraftAmbiguity	Prioritize potentially ambiguous draft clauses for human review	Revision-aware corpus, AI-assisted operational labels, grouped ML/LM evaluation	Separates corpus/alignment coverage from detector evaluation and keeps future evidence outside prediction features

Chapter 3

Methodology

3.1 Revision-Aware Study Pipeline

Figure 3.1 summarizes the pipeline. Every arrow crosses an artifact boundary. Raw documents and metadata remain unchanged; derived directories are versioned and immutable. Manifests record hashes of inputs, code, model snapshots, split assignments, and outputs.

3.1.1 Collection and structural parsing

The curated corpus covers application, DNS, routing, security, and transport standards. Twelve RFCs and all available histories for ten modern draft families are stored; OSPFv2 and BGP-4 are RFC-only because their historical drafts are not available through the same archive. The audit reports 253 expected and present draft revisions, 12 expected and present RFCs, and 31 verified technical errata.

The parser accepts both Datatracker plaintext and XML2RFC. It emits a common document–section–block tree while retaining original text, section paths, source locations, reference targets, tables, artwork, and formal syntax. A deterministic clause layer conservatively separates prose and list items. It recognizes uppercase BCP 14 keywords while guarding quoted text, references, abbreviations, version numbers, URLs, and condition–requirement pairs. Each clause receives a stable identifier and one of Normative, Contextual, Excluded, or Manual Review. This stage identifies likely requirements; it does not decide ambiguity.

3.1.2 Local RFC-review prototype

The project includes a local browser interface that accepts rendered RFC plaintext and applies the saved engineered-feature plus TF–IDF logistic-regression model at its saved threshold only to clauses the evidence-first parser classifies as normative, based on explicit uppercase BCP 14 requirement keywords and its quotation, reference, and terminology

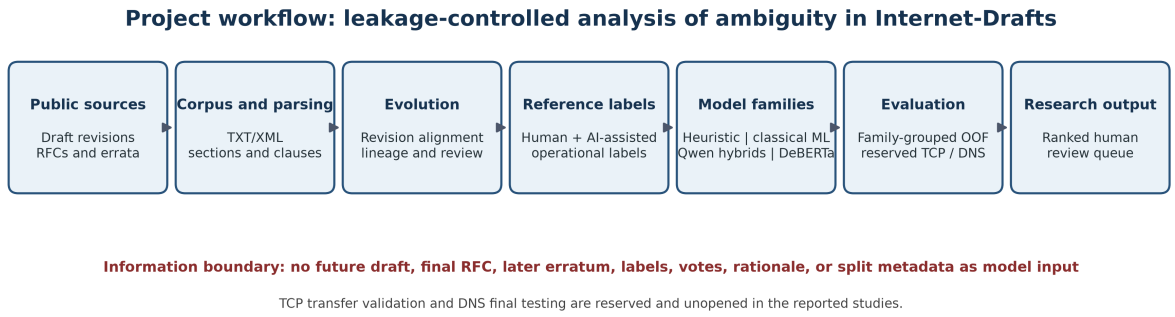


Figure 3.1: The intended DraftAmbiguity research pipeline. Future revisions, final-RFC text, later errata, votes, rationales, family identifiers, and split names are excluded from model features. The proposal reserved TCP and DNS, but the latest all-family classical rerun subsequently consumed both families; the nine-family transformer and Qwen studies still exclude them.

exclusions. Lowercase-modal and section-context candidates are not scored or presented for human normativity confirmation. The interface displays and exports only positive ambiguity predictions, and optional local Qwen explanations are restricted to those results. Uploaded document text remains local and is not sent to a remote service. This conservative gate can omit genuine normative clauses that do not use explicit BCP 14 terms; scores are not calibrated probabilities, explanations are hypotheses rather than evidence, and predictions are not adjudicated ambiguity judgments.

3.1.3 Consecutive-revision alignment

For revisions $t-1$ and t, bidirectional retrieval first searches corresponding and nearby sections, followed by a bounded global fallback. Candidate score $s(i,j)$ combines semantic, lexical, structural, positional, block-type, and protected-token evidence:

$$s(i,j) = \sum_k w_k s_k(i,j), \tag{3.1}$$

where semantic and lexical similarity each receive weight 0.38 in the semantic pilot, structure 0.12, and position, block compatibility, and protected-token evidence 0.04 each. Distinctive exact anchors are locked only when supported by section structure. One-to-one components use bipartite assignment with unmatched choices. Competing split and merge candidates of at most three adjacent clauses use a mixed-integer program with non-overlap constraints. Moves require section correspondence or reliable anchor inversions; close alternatives remain Uncertain.

MiniLM embeddings are computed from direct tokenizer-ID windows, avoiding text reconstruction between tokenization stages. The TLS coverage pilot aligns 30 documents across 29 consecutive pairs: 52,137 clause-version occurrences produce 52,953 relation-

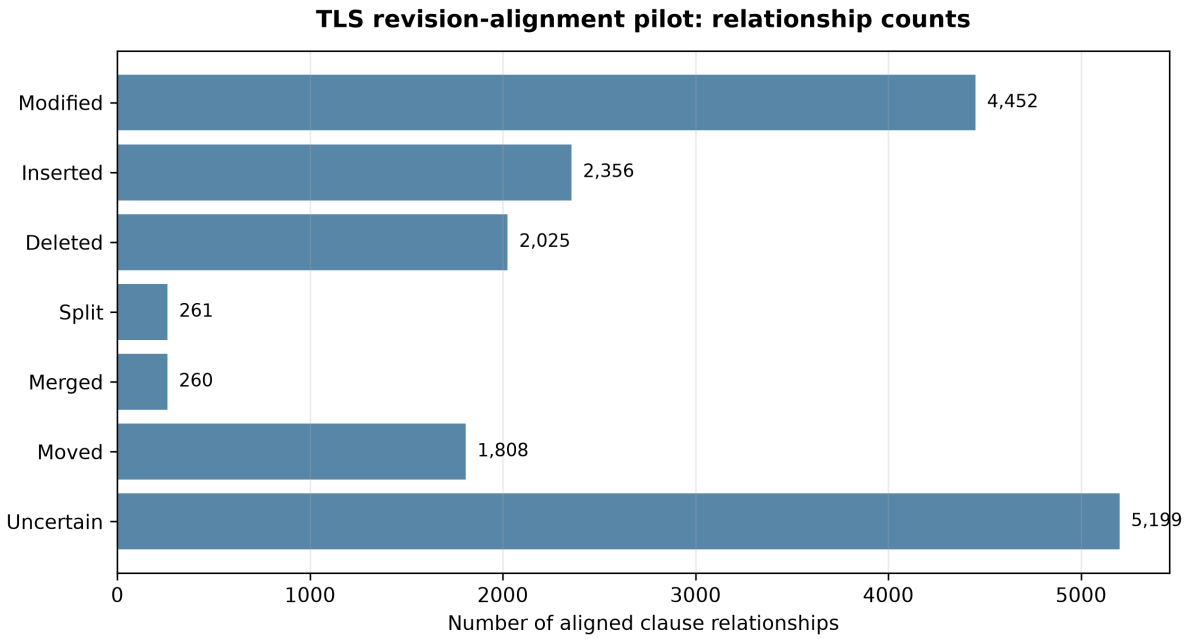


Figure 3.2: Relation counts from the TLS alignment coverage pilot. Uncertain links are retained for review; counts are not accuracy estimates.

ships, including 4,452 modified, 2,356 inserted, 2,025 deleted, 261 split, 260 merged, 1,808 moved, and 5,199 uncertain relationships. Fig. 3.2 visualizes these counts. They establish pipeline coverage, not alignment accuracy: the stratified 150-link review sample has not been adjudicated, so link precision and recall are intentionally unreported. Revision links are not model features in the transformer experiment reported below.

3.1.4 Temporal and leakage controls

For a clause in revision t, revision features may depend only on revisions $\leq t$. Later drafts, RFC text, later errata, undated evidence, annotation metadata, and model votes are rejected by a feature guard. RFC endpoints are available for offline correspondence research but not for draft-stage prediction. Family grouping blocks related clauses from crossing train and evaluation roles. Exact normalized cross-partition target overlap is zero in the current artifact. Fitting-side exclusions additionally cover near duplicates, shared lineages, BCP 14 skeleton conflicts, and target-as-context reproduction.

3.2 Reference Labels and Partitions

3.2.1 Operational annotation reference

Candidate clauses were assembled from eleven draft-family review sets after lineage deduplication. Three local models (Gemma 3 27B, GPT-OSS 20B, and Qwen3 32B) inde-

pendently supplied binary or uncertain decisions. Two agreeing valid votes form the consensus; unanimous consensus is marked high-confidence and a two-of-three decision medium-confidence. Rows without a strict two-vote majority remain uncertain rather than being forced into the negative class. The training export excludes uncertain candidates, rationales, later-revision evidence, and errata-derived features.

The latest matched classical/heuristic comparison uses all 2,425 resolved rows in `final_annotation_ready.csv`: 160 positive and 2,265 negative clauses across eleven families. This rerun includes TCP and DNS and therefore supersedes the earlier 2,406-row classical result still described in the revision-feature summary markdown; that summary is stale relative to the current metrics manifests. Earlier Qwen/hybrid studies use a frozen 2,424-row annotation-ready source and its 2,142-row, 126-positive nine-family development cohort, with TCP (131 rows, 5 positives) and DNS SVCB/HTTPS (151 rows, 17 positives) excluded. An earlier target-only transformer study used a hash-pinned subset of 2,125 rows (136 positives) after family and label-blind fitting exclusions. The updated `final_annotation_ready.csv` contains 2,055 model-agreement, 270 manual-conflict-verdict, and 100 `human_review_user_confirmed` records, including resolution of the previously unresolved `item_id=2161`. Its transformer development cohort contains 2,143 rows (138 positives) across nine families. The 100 reviewed cases were selected from historical OOF errors, so this updated cohort is error-selection biased and is not an independent or population-representative gold test. The updated DeBERTa follow-up compares focal loss, partial fine-tuning, and their rank ensemble on the same updated cohort; the reported analysis repair reuses the parent experiment's validated predictions and does not create independent evaluation evidence. Labels throughout this report are an *operational reference*, not independently double-annotated expert gold; metrics quantify agreement with a particular annotation artifact rather than universal ambiguity-detection accuracy.

3.2.2 Annotation disagreement audit

The annotation reference is noisy and model-dependent. In the earlier 2,546-row consensus snapshot, 435 of 455 positive rows (95.6%) had a 2–1 vote split; only 20 were unanimous positive. Among negative rows, 1,325 of 2,091 (63.4%) had a 2–1 split. This corresponds to an implied weighted positive share of 14.3%, compared with 17.9% unweighted. A later leave-one-model-out audit on vote-eligible subsets measured pairwise Cohen's κ of 0.113 (Gemma/GPT-OSS), 0.017 (Gemma/Qwen3), and 0.046 (GPT-OSS/Qwen3), despite raw agreement ranging from 0.370 to 0.792. Agreement-conditioned tiers were structurally extreme: the unanimous tier had perfect leave-one-out AP by construction, whereas the medium tier was highly imbalanced and often too small or single-class for stable estimates. The held-out-model vote APs were 0.0157 for Gemma, 0.1047 for GPT-

Table 3.1: Development cohort evaluated in the latest reviewed-label transformer study. Reserved TCP/DNS rows are not included in this experiment.

Role	Families	Rows	Yes	Use
Development	9	2,143	138	Five-fold family-isolated OOF

OSS, and 0.3825 for Qwen3 on their respective eligible subsets. Separately, the consensus-trained predictor's AP changed from 0.275 when scored against consensus to 0.737 against Gemma labels, 0.299 against GPT-OSS labels, and 0.028 against Qwen labels on the associated diagnostic snapshot. This target-sensitivity analysis is not a ranking of annotator quality; it reinforces that scores depend strongly on which model's decisions define the target. These audits do not identify which annotator is correct; they show that the consensus target itself is uncertain. A proposed vote-weighted model was not run because the pre-established stop rule triggered when most rows in both classes had weight 0.5.

3.2.3 Family-isolated partitions

The 11 labeled families are assigned by protocol family. The latest reviewed-label transformer study evaluates 2,143 development clauses from nine families (138 positives and 2,005 negatives; prevalence 6.4%). Its 100 human-confirmed records were selected from historical OOF errors, which limits population-level interpretation. TCP is reserved for a transfer check after the method is frozen, and DNS is reserved for one final test. Neither reserved partition was used for training, tuning, calibration, or analysis in the reported study.

Development uses five fixed outer folds. QUIC Transport, HTTP Semantics, and QUIC HTTP each form a held-out fold; IPv6, Cache-Status, HTTP/2, and TLS share one fold; QUIC Recovery and QUIC TLS share the remaining fold. Every fold assignment is fixed and hash-pinned. Exact, near-duplicate, lineage, BCP 14-skeleton, and target-as-context exclusions are applied to fitting designs.

3.2.4 Project-wide study map

Table 3.2 is the comparison key for the results that follow. It deliberately does not rank models across rows: annotation versions, candidate inclusion, prevalence, and validation protocols differ.

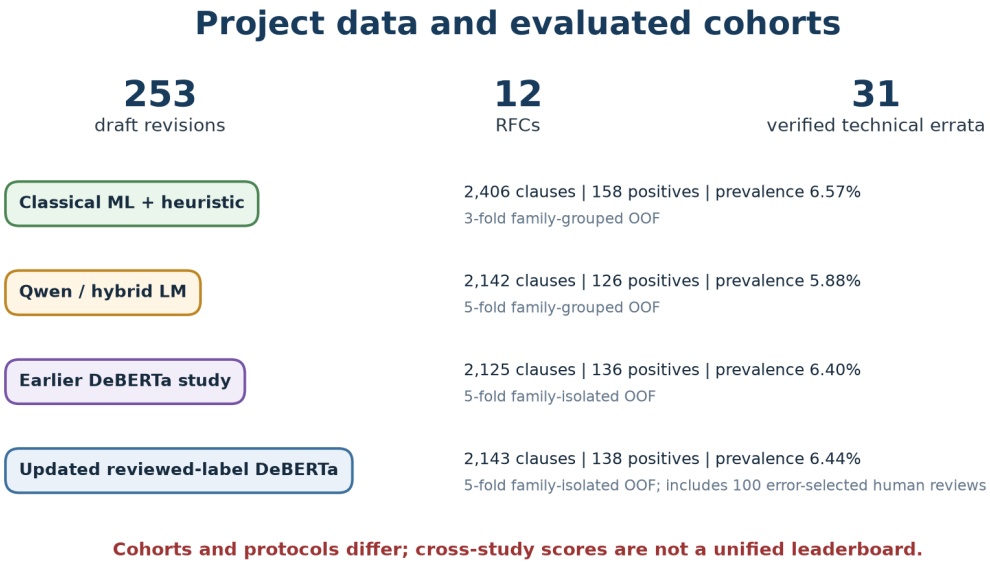


Figure 3.3: Corpus scale and principal evaluation cohorts. The updated reviewed-label DeBERTa comparison is shown separately from earlier classical, Qwen/hybrid, and transformer studies; only the three DeBERTa methods in Table 5.1 share the same updated cohort and folds.

3.3 Models and Experimental Protocol

3.3.1 Evaluation measures

For scores p_i and labels y_i, the primary ranking metric is average precision (AP), computed as the stepwise precision-weighted increase in recall. Repository reports historically call this "PR-AUC"; here we use the more precise name AP. Its no-skill reference is the positive prevalence, 0.064 in the evaluated development cohort. We also report ROC-AUC, precision, recall, F_1, F_2, balanced accuracy, confusion counts, and retrieval at fixed review budgets. Threshold-free rankings are primary because the appropriate operating point depends on reviewer capacity and the relative cost of missed ambiguity.

Uncertainty for the latest transformer study is estimated with 2,000 bootstrap resamples of the nine RFC families. These intervals represent sampling variation across the observed families; they do not include model-training variability, since the newest study has one seed.

3.3.2 DeBERTa variants and rank ensemble

Inputs contain only the target clause, with a maximum length of 256 tokens and deterministic head–tail truncation; the experiment reports no target truncation. Both models use five epochs, AdamW, learning rate 10^{-5}, weight decay 0.01, a linear schedule with 10% warmup, batch size 8, FP32, deterministic execution, and seed 2718. The focal model

uses $\gamma = 2$ and fitting-side class weights ($\alpha_0 = 1$, $\alpha_1 = N_-/N_+$):

$$\mathcal{L}_{\text{focal}} = -\alpha_y(1 - p_y)^\gamma \log p_y. \tag{3.2}$$

The partial model uses weighted cross-entropy and trains only layers 8–11, the pooler, and classifier.

For the ensemble, each method's empirical cumulative distribution function is fit on the corresponding fitting-side inner OOF scores. The two percentile ranks are averaged with equal weight and applied to the untouched outer-fold predictions. The resulting value is a ranking score, not a calibrated probability.

3.3.3 Classical machine learning and heuristic

The latest classical/heuristic comparison uses 2,425 rows, with 160 positives (6.60%) from 11 draft families. Models are evaluated using three-fold stratified group cross-validation, keeping draft-family groups intact; feature scaling, TF–IDF vocabulary, and feature selection are fitted within each training fold. The logistic-regression model uses 25 engineered clause features concatenated with word unigram/bigram TF–IDF, class-balanced L1 regularization, and $C = 0.5$. The complete recorded model grid also evaluates logistic regression, Linear SVM, and random forest with TF–IDF, frozen 384-dimensional MiniLM embeddings, TF–IDF plus embeddings, engineered-plus-TF–IDF inputs, and engineered-plus-TF–IDF with twelve label-free revision features. These revision features summarize the two preceding draft transitions (the target's $t-1 \to t$ change and, where available, the predecessor's $t-2 \to t-1$ change); matching is constrained to drafts available no later than the target revision. The full results are in Table 5.7. This latest all-family run includes TCP and DNS, so it is grouped OOF development evidence rather than the untouched reserved-family evaluation proposed originally.

The fixed heuristic is not trained on labels. It assigns hand-set points to conditional clauses lacking exception markers, deictic expressions, mixed BCP 14 modal levels, subjective terms, and exceptions; numeric bounds and explicit target references subtract points. The score is clipped to $[0, 9]$, normalized by nine, and thresholded at 0.4. Its rules and threshold are fixed rather than fit per fold. We report average precision from OOF scores and threshold metrics at the recorded operating threshold; the heuristic is a transparent low-cost baseline, not a learned classifier.

3.3.4 Earlier family-held-out classical study

The repository also contains a distinct earlier study with a nine-family leave-one-family-out protocol, 2,540 evaluated clauses, and 17.8% positive prevalence. Its combined target TF–IDF/engineered baseline (E0) has pooled OOF AP 0.291 (95% bootstrap interval

0.260–0.335). Frozen MiniLM and BGE-small representations with engineered features reach AP 0.246 and 0.245, respectively. Simple one-revision predecessor signals yield AP 0.168 and 0.165. These results provide historical methodology context only; their cohort and labels differ from the latest 2,425-row ML/heuristic comparison and from the transformer/LM cohorts.

3.3.5 Language-model and hybrid study

We also report the repository's separate Qwen3-32B development study [15], which uses the frozen `final_annotation_ready.csv` label artifact: 2,142 clauses from nine families, including 126 positives (5.88%). It uses the five family-grouped outer folds defined for that artifact and is separate from the updated 2,143-row reviewed-label DeBERTa cohort. The model is a local quantized Qwen3 32B inference model; V1 and V2 update no model parameters. V1 compares zero-, four-, and eight-shot prompts, selecting demonstrations from fitting families and scoring the `yes`/`no` alternatives from native token log-probabilities. V2 tests stricter prompts with 8, 16, or 24 fold-safe demonstrations, including prevalence-aware class proportions, and estimates threshold operating points by cross-fitting. V3 combines frozen Qwen zero-shot scores with a nested, target-only word/character TF–IDF logistic classifier; its regularization and blend weight are chosen using inner-fold predictions. V4 combines the V3 score with frozen MiniLM semantic representations of the target and local context using the Sentence-BERT bi-encoder paradigm [16]. The semantic classifiers use class-balanced logistic regression ($C = 0.03$); the 0.80/0.10/0.10 V3/target/context combination was selected on development data. V3 and V4 are supervised hybrid models, not parameter-fine-tuned LLMs. An additional exploratory comparison maps each component's scores to empirical within-cohort percentile ranks, averages the ranks with equal weight, and evaluates only the shared label-compatible rows; neither weights nor a classification threshold are fitted. This comparison reuses OOF predictions from different label snapshots and is not an independent test. Results for the Qwen study and cross-model rank ensemble are reported in Section 5.7.

3.3.6 Nested operating points

Threshold selection is performed only on the fitting side. For each outer fold, three family-disjoint inner folds estimate method-specific thresholds. The policy maximizes recall subject to minimum fitting precision in $\{0.20, 0.30, 0.50\}$, breaking ties by higher precision and then higher threshold. Infeasible targets remain missing; they are not replaced by an always-positive rule. Precision targets are objectives on inner data, not guarantees on a held-out family.

Table 3.2: Distinct project evaluation tracks and their principal development result. Values across rows are not head-to-head comparisons.

Track	Label artifact / cohort	Evaluation design	Principal result	Interpretation boundary
Earlier feature study	Consensus snapshot; 2,540 evaluated clauses, 453 positives	Nine-family leave-one-family-out (LOFO)	Target TF–IDF + engineered features: AP 0.291	Candidate-enriched cohort; distinct all-eleven-family E0 AP is 0.293
Classical ML and heuristic	`final_annotation_ready.csv`; 2,425 clauses, 160 positives	Five-fold stratified family-grouped OOF	Engineered + TF–IDF Linear SVM: AP 0.3584; fixed heuristic: 0.0873	Latest all-family rerun; includes the proposed TCP/DNS holdouts
Qwen and hybrids	`final_annotation_ready.csv`; 2,142 clauses, 126 positives	Five family-grouped outer folds; nested hybrids	Eight-shot Qwen AP 0.1960; V4 hybrid AP 0.3252	V3/V4 are adaptive supervised development hybrids; TCP/DNS untouched
Original-annotation DeBERTa	Dataset A; 2,142 clauses, 45 positives	Five family folds; three seeds	Weighted CE mean-fold AP 0.1322; pooled OOF AP 0.0603	2.1% prevalence; threshold 0.5 retrieves no positives
Annotation-ready DeBERTa pilot/calibration	Separate 2,142-clause, 126-positive artifact	One-seed pilot; later nested five-fold calibration	Pilot AP 0.2332; nested ECDF ensemble AP 0.3715	Distinct runs/protocols; results are development-only
Transformer weighting/context ablations	Earlier consensus development artifact	Frozen five-fold grouped design; three seeds for reported ablations	Class weighting: mean-fold AP 0.3022; hard target-only selected over soft/context conditions	DeBERTa-v3-small; its data and selection rule differ from the final target-clause study
Earlier target-clause transformer	Hash-pinned earlier subset; 2,125 clauses, 136 positives	Five-fold family-isolated OOF; nested fitting-side thresholds	Focal/partial/rank ensemble AP: 0.3054/0.3262/0.3711	DeBERTa-v3-base; one seed; retained as a separate historical result
Cross-model rank ensemble	Label-compatible subset; 2,097 rows, 116 positives	Fixed equal-weight within-cohort percentile-rank averaging	V4 + DeBERTa + classical ML: AP 0.4619	Exploratory pooled OOF; ML labels predate current source; not an independent comparison
Reviewed-label DeBERTa follow-up	Updated source; 2,143 development rows, 138 positives	Five-fold family-isolated OOF; parent predictions validated and analysis repaired	Focal/partial/rank ensemble AP: 0.5823/0.5761/0.6736	Same-cohort comparison within this study; one seed; 100 human-confirmed cases were error-selected; not an independent test

Chapter 4

Design and Implementation

4.1 System Architecture

The implementation follows a staged artifact architecture rather than a single end-to-end model. This makes temporal boundaries inspectable and allows each research claim to be traced to a specific dataset, model, and evaluation protocol. Figure 3.1 shows the data flow; Table 4.1 maps the principal components to their responsibilities and outputs.

4.2 Revision-Feature Implementation

The current classical experiment adds twelve compact numerical features to the engineered-plus-TF–IDF representation. H0 describes the immediate $t-1 \rightarrow t$ transition and H1 describes $t-2 \rightarrow t-1$ after following the H0 predecessor. Each tier includes predecessor availability; unchanged, modified, and moved indicators; and match-based and lexical change distances. The feature loader reads only `item_id`, `record_id`, and `family_id` from the annotation identity key. It rejects duplicate identities, unexpected matcher versions, and document mismatches. Labels, candidate-source fields, votes, rationales, future drafts, final-RFC endpoints, and errata are excluded. Numerical scaling is fit separately on each training fold, and every final manifest records the feature schema plus hashes of all matcher caches.

4.3 Model and Artifact Interfaces

The classical runners share one feature-construction interface so that logistic regression, Linear SVM, and random forest receive identically ordered inputs. Every experiment writes fold metrics, OOF scores, a metrics summary, a provenance manifest, raw revision features when used, and a final development model. Plot generation discovers saved metrics, checks cohort compatibility, and compares only runs with the same row count,

Table 4.1: Principal implemented components of DraftAmbiguity.

Component	Responsibility	Main output
Corpus builder	Acquire versioned drafts, RFC endpoints, metadata, and verified errata while preserving raw sources	Versioned document store and corpus manifest
Structural parser	Normalize XML/plaintext into documents, sections, blocks, and conservative clauses	Stable clause identifiers, source locations, context, and reference fields
Revision matcher	Match clauses to predecessors using section constraints, lexical similarity, and frozen MiniLM similarity	H0/H1 correspondence, relation type, and change distance
Annotation pipeline	Construct review candidates, retain provenance, and consolidate the operational binary reference	Frozen annotation snapshots, disagreement audits, and family-grouped folds
Model runners	Fit heuristic, classical, transformer, prompting, and hybrid variants under experiment-specific controls	OOF predictions, metrics, manifests, and final development models
Review application	Parse an RFC, score eligible clauses, filter/rank results, and optionally request a local explanation	Human-review queue with clause context and provenance

positive count, and fold composition. The report uses the OOF artifacts for evaluation; final models fit on all development rows are retained only for downstream prototype use.

4.4 Local Review Prototype

The local interface is a human-in-the-loop inspection tool. It parses a selected RFC, applies the saved classical model to eligible BCP 14 clauses, and presents ranked clauses with surrounding text and filtering controls. Optional local-Qwen explanations are requested only for selected results and do not alter the classifier score. This separation prevents generated explanations from silently becoming prediction features. The interface does not implement an autonomous publication gate, and the current threshold is not deployment-calibrated.

4.5 Implementation Choices and Reproducibility

Python, pandas, SciPy, scikit-learn, PyTorch, Hugging Face Transformers, and sentence-transformers provide the main implementation stack. Fixed seeds, immutable input manifests, SHA-256 hashes, family-level fold assignments, and serialized preprocessing objects support repeatability. Cached local-model outputs reduce nondeterministic or costly reruns. The repository nevertheless does not constitute a completely self-contained public artifact: private annotation exports, model weights, and some large local runs are intentionally absent, and a fresh clone cannot reproduce every table without those inputs.

Chapter 5

Results and Evaluation

5.1 Updated Reviewed-Label Transformer Results

Table 5.1 and Fig. 5.1 report five-fold family-isolated OOF estimates for the updated 2,143-row cohort. All three methods use the same labels, rows, and outer folds. The equal-weight ECDF ensemble has the highest observed AP (0.6736), approximately 10.5 times the 0.0644 prevalence; focal loss has AP 0.5823 and partial fine-tuning 0.5761. The 95% family-cluster bootstrap intervals overlap. With only nine family clusters and one training seed, these results are descriptive and do not establish a statistically reliable winner or quantify seed variation. The 100 human-confirmed records were selected from historical OOF errors, so this cohort is not an unbiased sample of the target population.

At the diagnostic 0.5 threshold, focal loss has precision 0.585 and recall 0.623; partial fine-tuning has precision 0.535 and recall 0.609; the ensemble has precision 0.584 and recall 0.630. The ensemble value is an ECDF rank, not a calibrated probability, and 0.5 is not a deployment-validated threshold.

The top-20 and top-50 results favor the ensemble on this pooled OOF ranking, but they are small-budget point estimates from a cohort enriched by historical model errors. They are not calibrated confidence estimates or expected production yields.

5.2 Additional Transformer Pilots, Calibration, and Protocol History

The project contains multiple earlier transformer evaluations in addition to the updated reviewed-label results above. They are reported here as distinct experiments rather than merged estimates.

An initial Dataset A run evaluated DeBERTa-v3-base on 2,142 rows with 45 positives (2.10% prevalence), across five family folds and three seeds. Standard cross-entropy

Table 5.1: Five-fold family-isolated OOF ranking results on the updated 2,143-row development cohort (138 positives). The 95% intervals resample nine families.

Method	AP	ROC-AUC	95% family CI (AP)	F_1 at 0.5
Focal loss	0.5823	0.8815	0.5355–0.6900	0.6035
Partial fine-tuning	0.5761	0.8924	0.5210–0.6982	0.5695
Equal-weight ECDF ensemble	**0.6736**	**0.9118**	0.5920–0.7641	**0.6063**
Prevalence baseline	0.0644	–	–	–

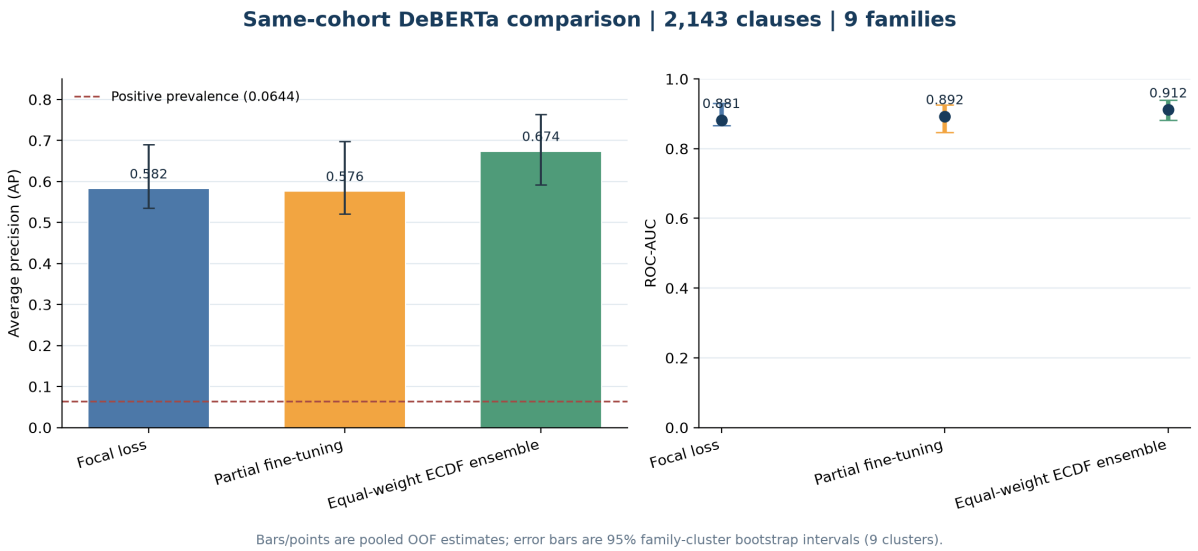


Figure 5.1: Average precision and ROC-AUC for the three methods evaluated on the same updated 2,143-row cohort. Error bars are 95% family-cluster bootstrap intervals; overlap and the small number of families preclude a superiority claim.

yielded mean-fold AP 0.0777 and mean pooled OOF AP 0.0392; weighted cross-entropy yielded 0.1322 and 0.0603, respectively. Neither predicted a positive at threshold 0.5, despite 0.979 accuracy. This result illustrates why accuracy is unsuitable for this imbalance and why this artifact cannot be conflated with the later 126-positive label snapshot.

On the earlier annotation-ready artifact with 2,142 rows and 126 positives, a one-seed weighted-cross-entropy pilot achieved AP 0.2332 (95% family-cluster interval 0.2020–0.3371) and ROC-AUC 0.7286. Its threshold sweep was descriptive and post hoc. A later nested study on that artifact compared focal loss (AP 0.3295), partial fine-tuning (0.3008), and an equal-weight fitting-side empirical-CDF ensemble (AP 0.3715; ROC-AUC 0.8125; 95% family-cluster interval 0.2551–0.5980). At a fitting-side precision objective of 0.30, the ensemble's pooled OOF precision and recall were 0.3273 and 0.4286; at fixed global review budgets, it retrieved 31 positives in the top 50 and 43 in the top 100. The corresponding top-budget comparison is shown in Table 5.3. These intervals resample only nine families and do not quantify seed variation. These pilot and nested-calibration results use a different annotation artifact and selection protocol from the updated reviewed-label

Table 5.2: Pooled OOF retrieval at fixed review budgets on the updated reviewed-label cohort. Top-20 and top-50 estimates are variable and reference-based.

Method	Positives@20	Precision@20	Positives@50	Precision@50
Focal loss	15	0.75	40	0.80
Partial fine-tuning	17	0.85	36	0.72
Equal-weight ECDF ensemble	**19**	**0.95**	**46**	**0.92**

Table 5.3: Fixed-budget retrieval for the separate 2,142-row annotation-ready DeBERTa calibration cohort. Counts are OOF positives against the operational reference, not independent test yields.

Method	Positives in top 50	Precision@50	Positives in top 100	Precision@100
Full fine-tuning, weighted CE	18	0.36	32	0.32
Focal loss	28	0.56	38	0.38
Partial fine-tuning	26	0.52	43	0.43
Equal-weight ECDF ensemble	31	0.62	43	0.43

results in Table 5.1; neither replaces the other.

An audit of an early learning-rate screen found that outer-fold scores had been used for checkpoint selection and then reported on those same folds. Those estimates were retired. The corrected nested screen on a separate 2,253-row, 395-positive consensus training artifact selected learning rate 10^{-5}, obtaining mean-fold AP 0.2817 and pooled AP 0.2563. For completeness, the corrected seed-1729 screen produced the following mean-fold/pooled AP pairs: 10^{-5}, 0.2817/0.2563; 2×10^{-5}, 0.2683/0.2474; 3×10^{-5}, 0.2600/0.2578; and TF–IDF, 0.2823/0.2712. A second-seed rerun of the top two learning rates obtained mean-fold AP 0.3153 for 10^{-5} and 0.2865 for 2×10^{-5}; the selected-rate pooled AP was 0.2774 versus TF–IDF 0.2712, with a section-cluster interval for the difference that included zero.

The subsequent feasibility audit estimated the referral fractions needed to meet 80%, 90%, and 95% recall. Table 5.4 reports its principal methods; the prevalence-only score requires referring every row, while the oracle row is a non-operational lower bound. The two-seed DeBERTa average had mean fold-to-fold seed AP standard deviation 0.0237 (maximum 0.0459). These internal development estimates are not deployment guarantees and do not establish a reliable transformer advantage over TF–IDF.

The project also attempted annotation v2 with a new Qwen3.5 27B, Gemma 4 31B, and GPT-OSS 120B panel on a planned 2,312-candidate development sample. Hardware preflight blocked all 72 planned cells: available system RAM fell below the reserved safety margin for the 27B and 31B probes, and the 120B model's conservative peak estimate exceeded available memory. Only two synthetic schema probes were sent; no development clause was sent to these models, and there are no v2 labels or agreement results. The separate GPT-OSS 120B full-run and smoke directories likewise contain no

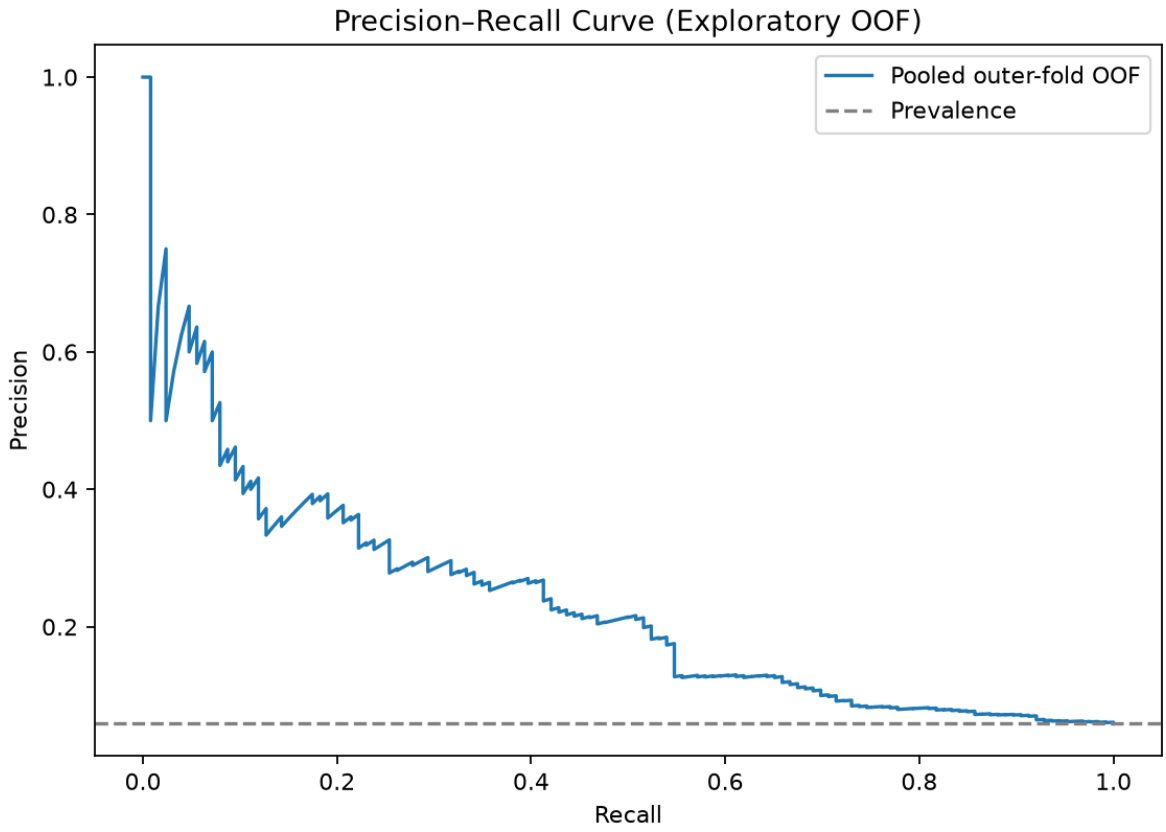


Figure 5.2: Pooled outer-fold precision–recall curve for the one-seed annotation-ready DeBERTa pilot (2,142 rows, 126 positives). This exploratory pilot is separate from the later nested calibration, the earlier 2,125-row target-only study, and the updated reviewed-label follow-up.

completed aggregate analysis. These lines are not model results. A recall-improvement report records the same 0.2332 weighted-cross-entropy pilot value, but its planned nested-threshold, context, focal-loss, and partial-freezing stages were skipped for time and cannot support conclusions about those methods.

5.3 Earlier Transformer Ablations

The repository also contains two protocol-frozen DeBERTa-v3-small ablation tracks on earlier consensus artifacts [17]. These results are included to document the project-wide model comparisons, not pooled with either the earlier 2,125-row or updated 2,143-row target-clause experiment. Step 3 compared unweighted, class, confidence, and combined weighting over five fixed family folds and three seeds. Class weighting was selected by the frozen seed-averaged mean-fold criterion (0.3022); its 0.0067 advantage over unweighted training had paired-fold standard error 0.0055. All section-cluster intervals for the pooled DeBERTa–TF–IDF differences included zero, so the hypothesis that DeBERTa reliably outperformed the matched surface-text comparator remained unsupported. At a devel-

Table 5.4: Estimated fraction of clauses referred to reach target recall in the earlier feasibility audit. Point estimates are development-only and thresholds are not deployment-calibrated.

Method	80% recall	90% recall	95% recall
TF–IDF, unweighted	0.611	0.755	0.834
TF–IDF, class-weighted	0.611	0.751	0.846
DeBERTa, seed 1729	0.685	0.851	0.916
DeBERTa, seed 2718	0.702	0.841	0.885
DeBERTa, two-seed logit average	0.638	0.819	0.891
Prevalence-only score	1.000	1.000	1.000
Perfect-ranking oracle	0.141	0.159	0.168

Table 5.5: Earlier Step 3 weighting ablation. Mean-fold AP is the frozen selection statistic; pooled AP weights rows rather than folds.

Weighting	DeBERTa mean-fold AP	DeBERTa pooled AP
Unweighted	0.2955	0.2663
Class	**0.3022**	0.2874
Confidence	0.2907	0.2708
Combined	0.2949	0.3016

opment threshold chosen to attain at least 90% recall, the selected DeBERTa condition referred 85.8% of rows, versus 73.4% for the selected TF–IDF condition; these are uncalibrated OOF routing estimates, not service-level guarantees.

Step 4 used a frozen three-seed factorial comparison of hard versus soft targets and target-only versus local-context inputs. Under the pre-established mean-fold selection rule, hard target-only (0.3022) was selected over soft target-only (0.3014); the 0.0008 difference was much smaller than its paired-fold standard error (0.0063). Adding local context reduced mean-fold AP for both hard and soft labels. Although soft targets yielded a lower estimated referral rate at 90% recall (73.8% versus 85.8% for hard target-only), this secondary routing result did not replace the frozen primary selection. A matched nested TF–IDF comparison was directionally lower (mean-fold AP 0.2763) but its paired section-cluster interval for pooled AP difference included zero. No robust superiority claim is supported.

5.4 Classical ML and Heuristic Results

Table 5.7 and Fig. 5.4 report the same latest 2,425-row, 11-family grouped-OOF cohort (160 positives; prevalence 6.60%). Linear SVM with engineered features plus TF–IDF

Table 5.6: Earlier Step 4 hard/soft target and context factorial. Each condition is from a separate earlier consensus artifact, not the final target-clause cohort.

Condition	Mean-fold AP	Pooled AP	Referral at 90% recall
Hard target only	**0.3022**	0.2874	0.858
Soft target only	0.3014	0.2997	0.738
Hard target + context	0.2652	0.2779	0.797
Soft target + context	0.2735	0.2903	0.755

Table 5.7: Complete matched classical ML and fixed-heuristic OOF grid on the latest 2,425-row cohort (160 positives). All threshold metrics use the recorded 0.5 threshold; AP and ROC-AUC are threshold-free.

Model	Features	AP	ROC-AUC	Precision	Recall	F_1
Logistic regression	Engineered + TF–IDF	0.3265	0.8082	0.1902	0.7063	0.2997
Logistic regression	Engineered + TF–IDF + revision	0.3185	0.8067	0.1902	0.7063	0.2997
Linear SVM	Engineered + TF–IDF	**0.3584**	0.8070	0.4545	0.3125	0.3704
Linear SVM	Engineered + TF–IDF + revision	0.3572	0.8064	0.4522	0.3250	0.3782
Random forest	Engineered + TF–IDF	0.3363	0.8197	0.9000	0.0563	0.1059
Random forest	Engineered + TF–IDF + revision	0.3348	0.8126	0.9231	0.0750	0.1387
Logistic regression	TF–IDF	0.1967	0.7720	0.1594	0.6875	0.2588
Linear SVM	TF–IDF	0.2574	0.7720	0.4306	0.1938	0.2672
Random forest	TF–IDF	0.2294	0.7752	1.0000	0.0250	0.0488
Logistic regression	MiniLM embeddings	0.1439	0.6708	0.1699	0.3250	0.2232
Linear SVM	MiniLM embeddings	0.1231	0.6272	0.1673	0.2938	0.2132
Random forest	MiniLM embeddings	0.1984	0.7430	1.0000	0.0063	0.0124
Logistic regression	TF–IDF + MiniLM	0.1438	0.6708	0.1699	0.3250	0.2232
Linear SVM	TF–IDF + MiniLM	0.1308	0.6379	0.1736	0.2875	0.2165
Random forest	TF–IDF + MiniLM	0.2311	0.7654	1.0000	0.0063	0.0124
Fixed rule-based heuristic	Engineered rules	0.0873	0.6082	0.1190	0.0313	0.0495

attains the highest AP (0.3584), followed by random forest (0.3363) and logistic regression (0.3265) with the same feature family. The fixed heuristic has AP 0.0873, only modestly above prevalence and substantially below the learned text-feature models. At the recorded 0.5 threshold, the heuristic has precision 0.119 and recall 0.031; logistic regression has recall 0.706 and precision 0.190, whereas the SVM has recall 0.313 and precision 0.455. Adding revision features lowers AP from 0.3265 to 0.3185 for logistic regression, from 0.3584 to 0.3572 for SVM, and from 0.3363 to 0.3348 for random forest. The rerun therefore does not support a ranking improvement from these features. These are exploratory estimates for this cohort, which includes the proposed TCP and DNS holdouts, and should not be compared directly with the transformer or LM cohorts.

5.5 Earlier Family-Held-Out Classical Results

In the distinct earlier nine-family LOFO study, the target TF–IDF plus engineered-feature baseline reaches AP 0.291 (95% bootstrap interval 0.260–0.335) against 0.178 prevalence. MiniLM and BGE-small embeddings paired with engineered features reach 0.246 and

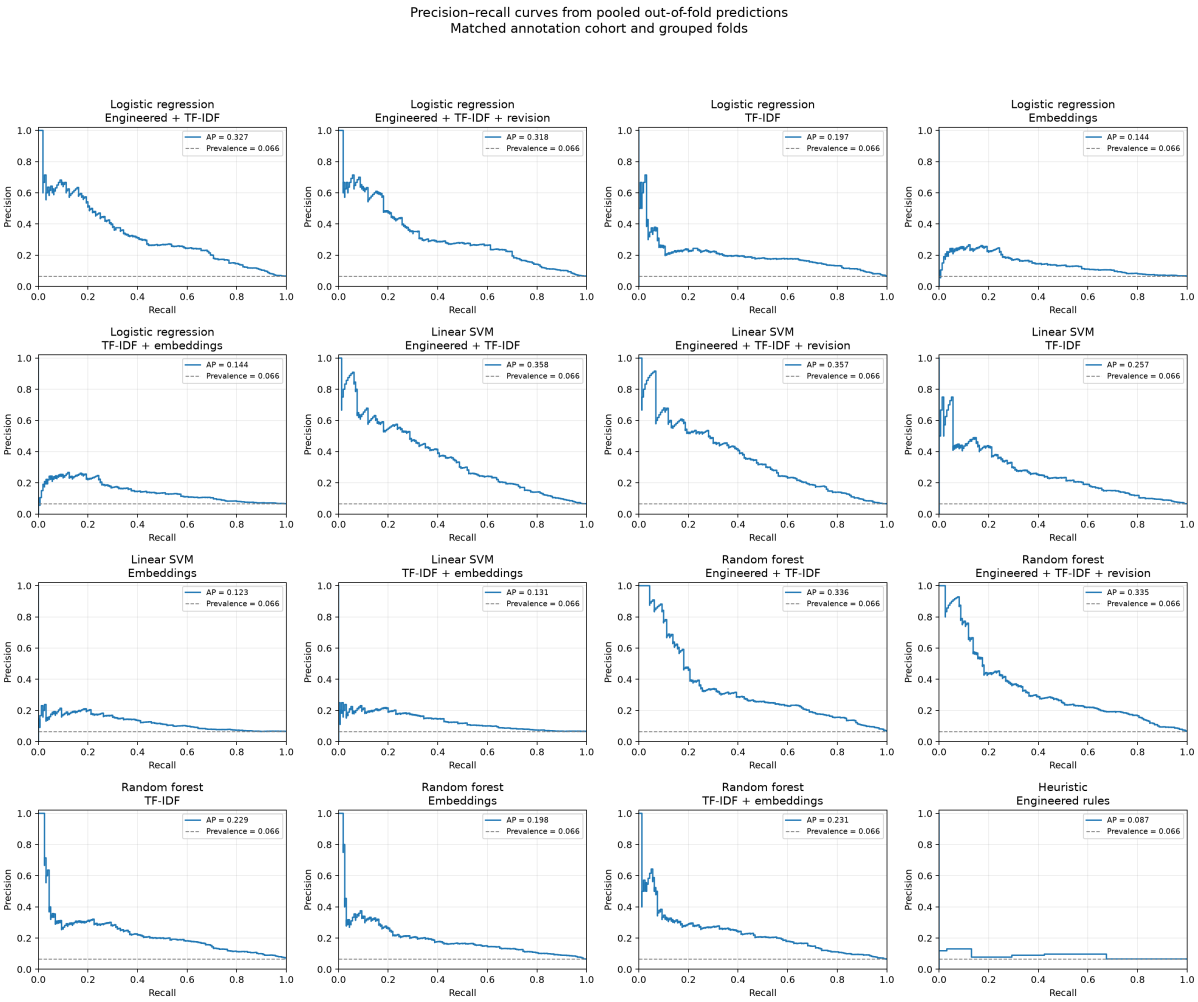


Figure 5.3: Pooled out-of-fold precision–recall curves for all 16 matched classical/heuristic configurations, including revision-augmented models. Each panel uses the latest 2,425-row cohort and grouped folds; the dashed line is prevalence.

0.245, respectively, while simple previous-revision signals remain at or below prevalence (0.168 and 0.165). Fig. 5.6 summarizes these results. This separate analysis suggests that sparse target text and engineered features were more useful than the tested frozen embeddings or simple churn features in that cohort; the comparison is not merged with the latest 2,425-row experiment.

5.6 Feature and Revision-Signal Comparisons

The direct revision ablation required by RQ2 holds the latest 2,425-row cohort, grouped folds, engineered features, TF–IDF representation, and classifier settings fixed, changing only the twelve past-only revision features. Table 5.8 and Fig. 5.7 show that pooled OOF AP decreases slightly for every classifier. Logistic-regression AP falls by 0.0081 with no change in F_2; SVM AP falls by 0.0012 while F_2 rises by 0.0110; random-forest AP falls by 0.0015 while F_2 rises by 0.0227. No probability-calibration measure was recorded for

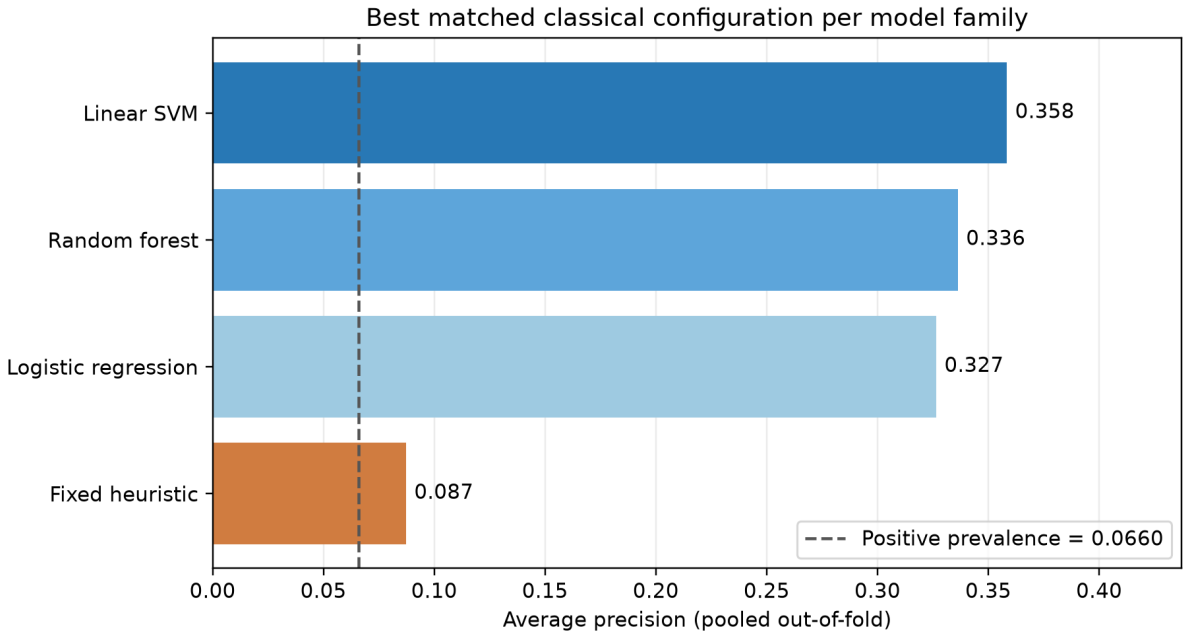


Figure 5.4: Average precision for classical classifiers and the fixed heuristic on their common grouped-OOF cohort. The dashed line marks cohort prevalence.

this ablation, and reference-count features were not isolated from the other engineered features. Consequently, the full revision/cross-reference/calibration claim in RQ2 is only partly tested and hypothesis H1 is not supported by ranking AP.

Table 5.8: Matched ablation of past-only revision features. Deltas are revision-augmented minus baseline scores.

Model	AP base	AP + revision	ΔAP	F_2 base	F_2 + revision	ΔF_2
Logistic regression	0.3265	0.3185	−0.0081	0.4579	0.4579	0.0000
Linear SVM	0.3584	0.3572	−0.0012	0.3333	0.3444	+0.0110
Random forest	0.3363	0.3348	−0.0015	0.0692	0.0919	+0.0227

Table 5.9 expands the earlier LOFO comparison beyond its headline baseline. The best frozen-embedding configuration (MiniLM or BGE-small plus engineered features) remained below the target TF–IDF-plus-engineered reference; adding previous/next-clause embeddings did not improve AP in this setup. A nested inner-CV embedding selector reached AP 0.232. Basic predecessor-only features produced AP 0.168 for the historical H1 pair and 0.165 for H0, near or below the 0.178 prevalence. These findings are specific to the recorded matching and feature pipeline: the predecessor matcher is one-to-one, has no split/merge handling, and could not be validated against retained row-level TLS links. The enriched candidate cohort itself was selected using future evidence, so these APs are not estimates over all draft clauses.

An exploratory previous-clause-text comparison raised AP only from 0.291 to 0.292 while F1 at threshold 0.5 fell from 0.365 to 0.325. The N1 score is only a selection-artifact

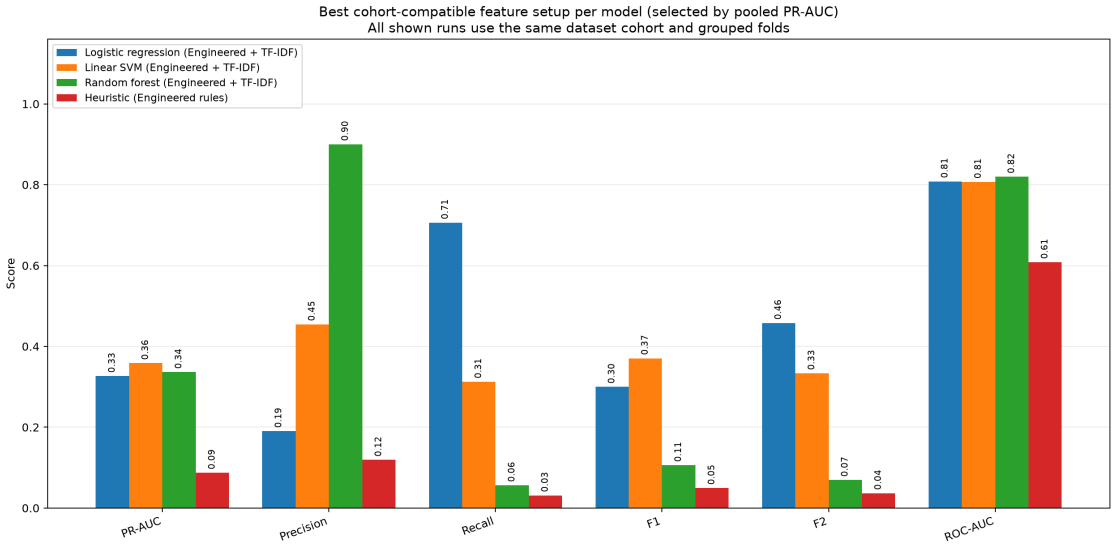


Figure 5.5: Threshold-free AP and ROC-AUC together with threshold-dependent precision, recall, F_1, and F_2 for the best feature setup per model in the matched classical/heuristic cohort. Threshold-dependent values use each run's recorded decision threshold and should not be interpreted as directly optimized operating points; Table 5.7 reports the complete model-feature grid.

diagnostic: its candidate-source field is not a permissible detector feature. The original all-eleven-family notebook reported E0 AP 0.293, engineered-only AP 0.275, target TF–IDF AP 0.273, LinearSVC TF–IDF AP 0.254, and random-forest engineered-only AP 0.240; this protocol is distinct from the later nine-family LOFO ablations. A separate fixed-split experiment with one 154-row test family reported AP 0.290 for engineered features, 0.325 for TF–IDF, and 0.315 for their combination; an earlier revision-history heuristic reported validation AP 0.332 and test AP 0.306 at a validation-selected threshold. These notebook-derived results lack the same grouped design and are exploratory, not merged into the family-held-out results.

5.7 Language-Model and Hybrid Results

Table 5.10 reports the prompting conditions and nested hybrid results on the separate 2,142-row label artifact. V1's highest pooled AP occurs at eight shots, but the frozen one-standard-error simplicity rule selects four shots because its mean-fold AP is within one paired-fold standard error of the eight-shot result. V2 compares strict 8-, 16-, and 24-shot prompts; 24-shot has the highest mean-fold AP, but the same one-standard-error rule selects 8-shot as the simpler condition. The V2 cross-fitted threshold rows demonstrate the precision–recall tradeoff: the F2-oriented 8-shot and 24-shot rules recall 84.1% and 80.2% of reference positives, respectively, at precision near 6.6%. AP is threshold-free; the tabulated precision, recall, F1, and balanced accuracy use each condition's recorded

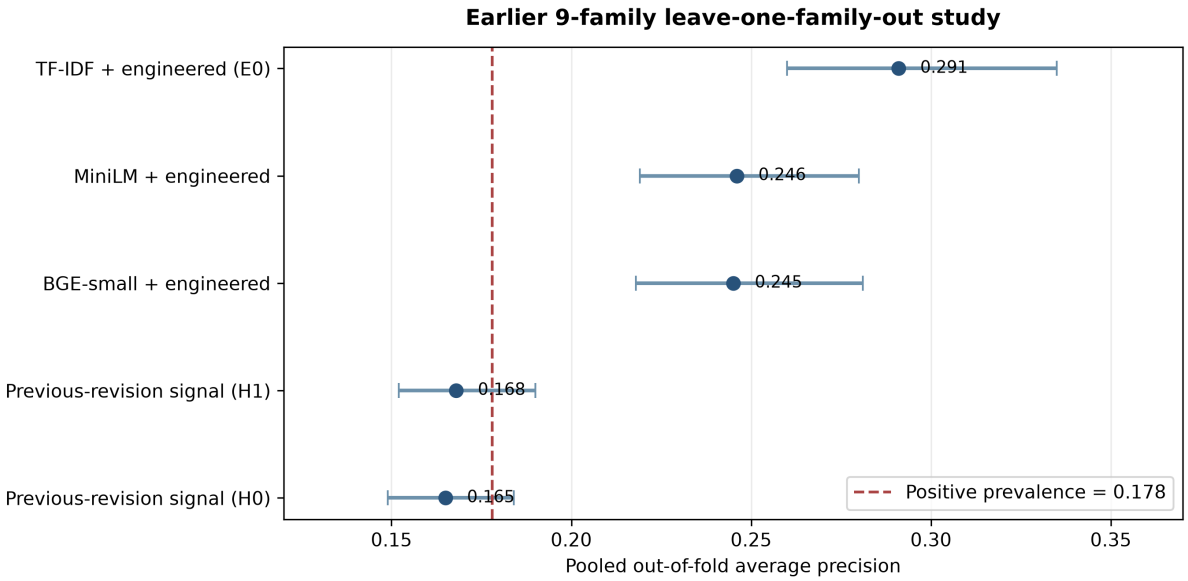


Figure 5.6: Earlier nine-family LOFO AP results with reported bootstrap intervals where available. The dashed line is this cohort's 0.178 positive prevalence.

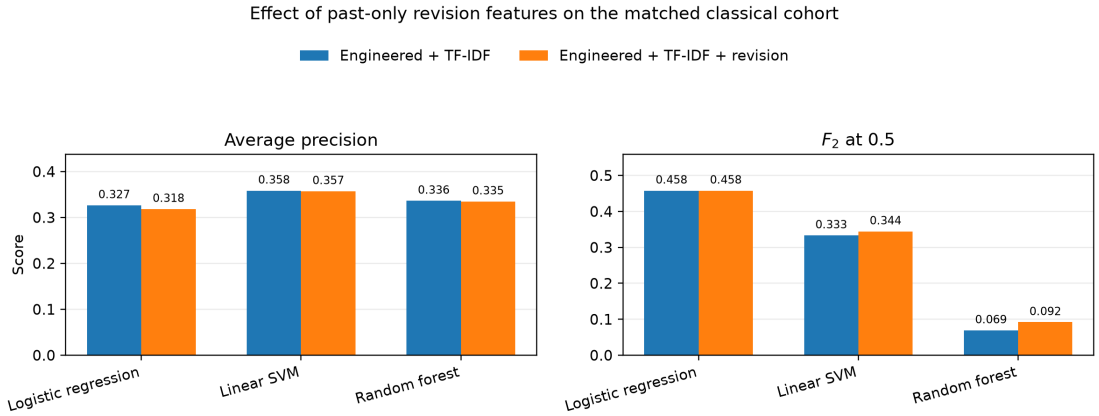


Figure 5.7: Average precision and F_2 before and after adding twelve leakage-controlled revision features. The mixed changes do not support a model-independent benefit from revision history.

raw or cross-fitted operating rule. V3 and V4 were developed after inspecting prior development results, so their improvements are adaptive development evidence rather than unbiased estimates of transfer performance.

Figure 5.8 plots the best reported AP from each version. The strict V2 prompt did not improve ranking over V1. The fitted lexical component in V3 increased AP over prompting alone, and V4's frozen semantic features provided a further development-set increase when combined with V3. MiniLM-only and naive Qwen/MiniLM combinations did not clearly exceed V3; V4 retained semantic features for complementary errors, not because the standalone encoder was a stronger detector. The operating-point results also reveal the V1 precision–recall tradeoff and V2's low-recall behavior. These results do not establish that LLMs outperform DeBERTa and must not be directly compared across the

Table 5.9: Selected feature and revision ablations from the nine-family LOFO cohort (2,540 rows; prevalence 0.178). All rows use the documented cohort and are not compared with other annotation snapshots.

Representation / signal	Model family	AP	95% bootstrap interval
Target TF–IDF + engineered (E0)	Logistic regression	**0.291**	0.260–0.335
Target-only MiniLM embeddings (E1)	Logistic regression	0.219	0.195–0.250
MiniLM embeddings + engineered (E2)	Logistic regression	0.246	0.219–0.280
Target, previous, and next MiniLM embeddings (E3)	Logistic regression	0.204	0.183–0.230
MiniLM context embeddings + engineered (E4)	Logistic regression	0.212	0.190–0.240
MiniLM full stack with target TF–IDF (E5)	Logistic regression	0.212	0.190–0.240
Target-only BGE-small embeddings (E1)	Logistic regression	0.224	0.199–0.256
BGE-small embeddings + engineered (E2)	Logistic regression	0.245	0.218–0.281
BGE-small target, previous, and next embeddings (E3)	Logistic regression	0.204	0.183–0.229
BGE-small context embeddings + engineered (E4)	Logistic regression	0.210	0.186–0.238
BGE-small full stack with target TF–IDF (E5)	Logistic regression	0.211	0.189–0.239
Nested inner-CV selected embedding	Logistic regression	0.232	0.207–0.265
Previous-revision predecessor signal (H1)	Logistic regression	0.168	0.152–0.190
Immediate predecessor signal (H0)	Logistic regression	0.165	0.149–0.184
Candidate-source-only N1 diagnostic	Logistic regression	0.179	0.160–0.199

two cohorts.

5.8 Final Three-Model Rank Ensemble

We additionally combine the V4 Qwen/MiniLM semantic ensemble, the DeBERTa-v3-base pilot, and the classical class-balanced logistic regression with engineered and TF–IDF features. The classical OOF artifact predates the current annotation snapshot: 17 current development records are absent, and 28 shared records have conflicting labels. We therefore exclude these records and evaluate all components on the same 2,097 label-compatible rows (116 positives; prevalence 0.0553). The equal-weight rank average obtains AP 0.4619 and ROC-AUC 0.8414, compared with AP 0.4567 for logistic regression alone, 0.3373 for V4, and 0.2517 for DeBERTa. V4 plus the classical model has AP 0.4582 and the highest ROC-AUC (0.8451) among the fixed combinations. No ensemble weights or decision threshold were tuned on these labels.

Figures 5.9 and 5.10 show ranking quality and the recall/review-budget trade-off for the components and fixed combinations. These pooled OOF curves are descriptive: the ML component was trained on an earlier annotation artifact, V4 and ensemble development used this cohort, and fold-specific scores are not deployment-calibrated. The result is not independent confirmation; the ML component should be retrained against the frozen final annotation source before further claims. Reserved TCP and sealed DNS results are not included.

Table 5.10: Qwen and hybrid development results on the separate 2,142-row, nine-family label artifact (126 positives; prevalence 0.0588). These results are not directly comparable to the updated reviewed-label transformer study.

Version	Condition / decision rule	AP	Precision	Recall	F_1	Balanced accuracy
V1	Qwen3 32B, zero-shot, raw	0.1513	0.0753	0.8492	0.1383	0.5987
V1	Qwen3 32B, 4-shot, raw (selected)	0.1772	0.0928	0.7937	0.1661	0.6543
V1	Qwen3 32B, 8-shot, raw	0.1960	0.0981	0.7778	0.1742	0.6654
V2	Qwen3 32B, strict 8-shot, raw	0.0869	0.0938	0.0714	0.0811	0.5141
V2	Qwen3 32B, strict 16-shot, raw	0.0828	0.1136	0.0397	0.0588	0.5102
V2	Qwen3 32B, strict 24-shot, raw	0.1035	0.1111	0.0635	0.0808	0.5159
V2	8-shot, cross-fitted F_1 threshold	0.0869	0.0691	0.4683	0.1204	0.5370
V2	8-shot, cross-fitted F_2 threshold	0.0869	0.0673	0.8413	0.1246	0.5561
V2	8-shot, cross-fitted balanced-accuracy threshold	0.0869	0.0642	0.6905	0.1174	0.5305
V2	16-shot, cross-fitted F_1 threshold	0.0828	0.0643	0.5714	0.1157	0.5260
V2	16-shot, cross-fitted F_2 threshold	0.0828	0.0605	0.6429	0.1106	0.5094
V2	16-shot, cross-fitted balanced-accuracy threshold	0.0828	0.0619	0.5952	0.1122	0.5159
V2	24-shot, cross-fitted F_1 threshold	0.1035	0.0684	0.4762	0.1196	0.5355
V2	24-shot, cross-fitted F_2 threshold	0.1035	0.0658	0.8016	0.1217	0.5454
V2	24-shot, cross-fitted balanced-accuracy threshold	0.1035	0.0678	0.5397	0.1205	0.5379
V3	TF–IDF + frozen Qwen, raw	0.2554	0.1737	0.5556	0.2647	0.6952
V3	TF–IDF + frozen Qwen, cross-fitted F_1	0.2554	0.2366	0.4206	0.3029	0.6679
V3	TF–IDF + frozen Qwen, cross-fitted F_2	0.2554	0.1538	0.6190	0.2464	0.7031
V3	TF–IDF + frozen Qwen, cross-fitted balanced-accuracy threshold	0.2554	0.1373	0.6984	0.2295	0.7121
V4	V3 + frozen MiniLM, raw	0.3252	0.2366	0.5238	0.3259	0.7091
V4	V3 + frozen MiniLM, cross-fitted F_1	0.3252	**0.3952**	0.3889	**0.3920**	0.6758
V4	V3 + frozen MiniLM, cross-fitted F_2	0.3252	0.1977	0.5397	0.2894	0.7014
V4	V3 + frozen MiniLM, cross-fitted balanced-accuracy threshold	0.3252	0.1440	0.7302	0.2405	0.7294

5.9 Review-Budget Analysis

Figure 5.11 plots results from the earlier 2,125-row target-only transformer study: its ensemble retrieves 26 of 50 clauses and 49 of 100. At that cohort's prevalence, random selection would retrieve 3.2 and 6.4 positives on average. For the updated reviewed-label cohort, Table 5.2 reports top-20 and top-50 retrieval. Both are exploratory prioritization diagnostics: scores were generated by fold-specific models, the ensemble score is not calibrated as a probability, and the reviewed cases were selected from historical OOF errors. Neither result should be interpreted as a promised deployment yield.

5.10 Threshold-Dependent Performance

The inner-fold policy selects thresholds to maximize recall subject to a minimum fitting-side precision target. With target precision 0.30, pooled outer OOF ensemble precision is 0.456, recall 0.382, and F_1 0.416. This does not mean a held-out family will achieve 0.30 precision: observed results vary by fold, and one fold's selected decision rule predicts every row positive for the focal model at this target. Table 5.11 summarizes the pooled results.

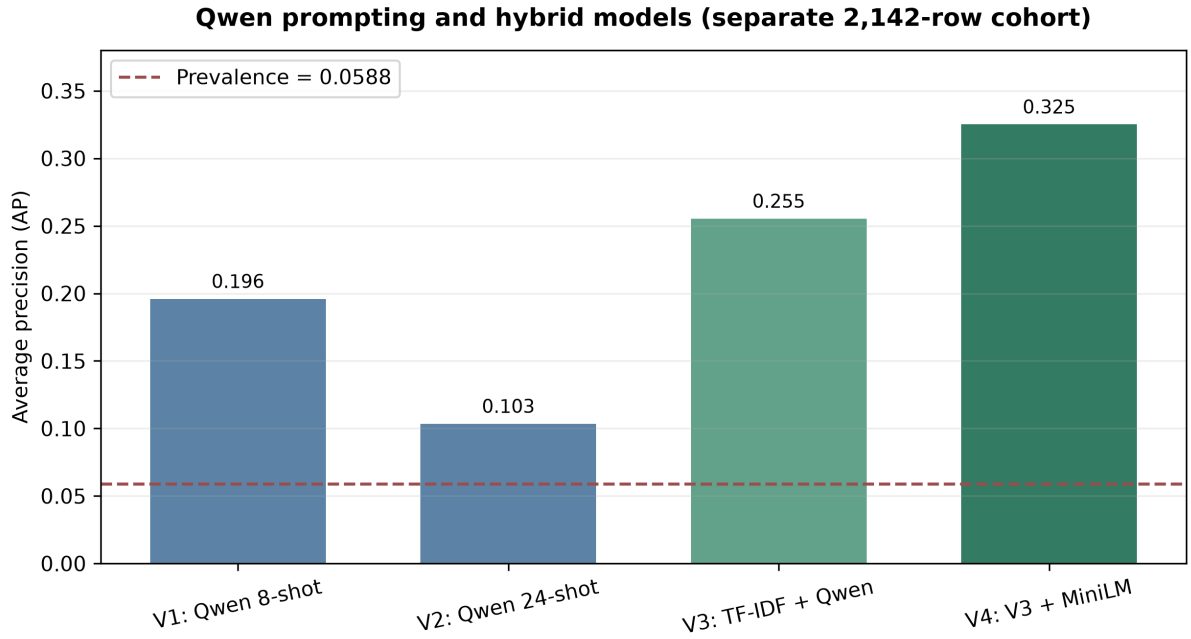


Figure 5.8: Best reported average precision for the separate Qwen/hybrid study. The dashed horizontal line denotes its positive prevalence (0.0588). V1/V2 are prompting conditions; V3/V4 include fitted classifiers.

Table 5.11: Pooled OOF operating points using thresholds selected on inner folds for a 0.30 fitting-side precision target in the updated reviewed-label study.

Method	Precision	Recall	F_1	F_2
Focal loss	0.384	0.732	0.504	0.620
Partial fine-tuning	**0.412**	0.710	**0.521**	0.620
ECDF ensemble	0.387	**0.732**	0.506	**0.621**

5.11 Family-Level Heterogeneity

The updated ensemble's family-specific AP ranges from 0.571 on QUIC TLS to 0.916 on TLS 1.3 among the larger families with positives. Several families are small: CacheStatus has five rows and two positives, with AP 1.0, and IPv6 has one negative row with undefined AP. These small and heterogeneous groups make pooled summaries insufficient on their own.

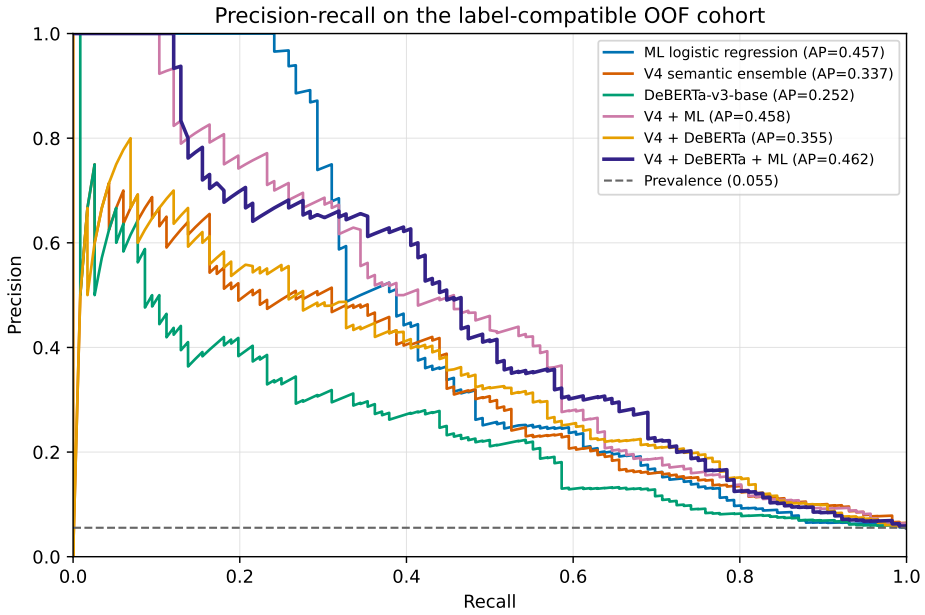


Figure 5.9: Pooled out-of-fold precision–recall curves for the three component models and fixed rank ensembles on the 2,097-row label-compatible subset. The dashed line is the subset prevalence (0.0553). Because the ML OOF artifact uses an earlier annotation snapshot, 17 unavailable rows and 28 label-conflicting rows are excluded. Development curves are exploratory, not independent test estimates.

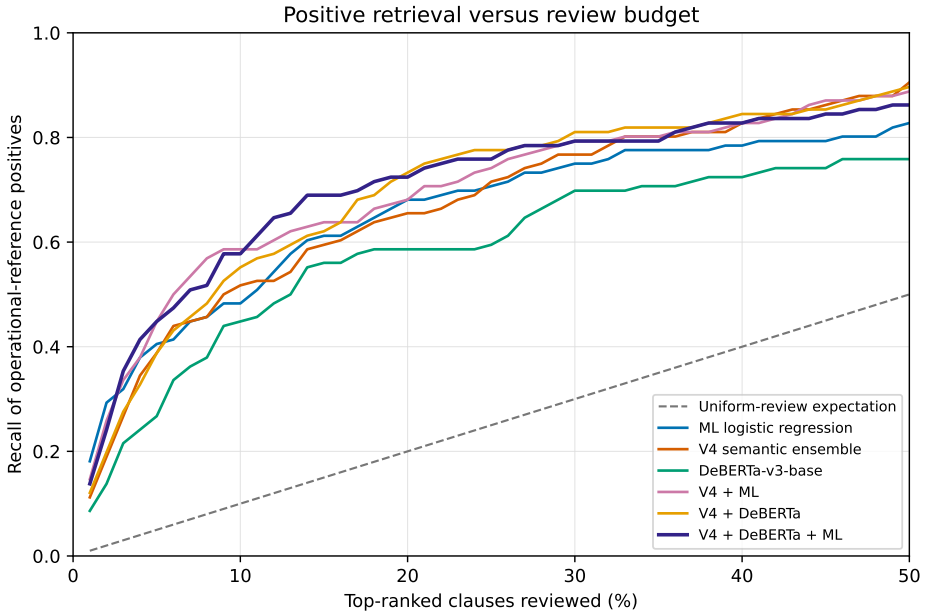


Figure 5.10: Recall of operational-reference positives among the highest-ranked clauses as the review budget increases. Uniform-review expectation is shown for reference. Scores pool fold-specific OOF predictions and are not calibrated deployment yields. The cohort is the 2,097-row label-compatible subset described in Fig. 5.9.

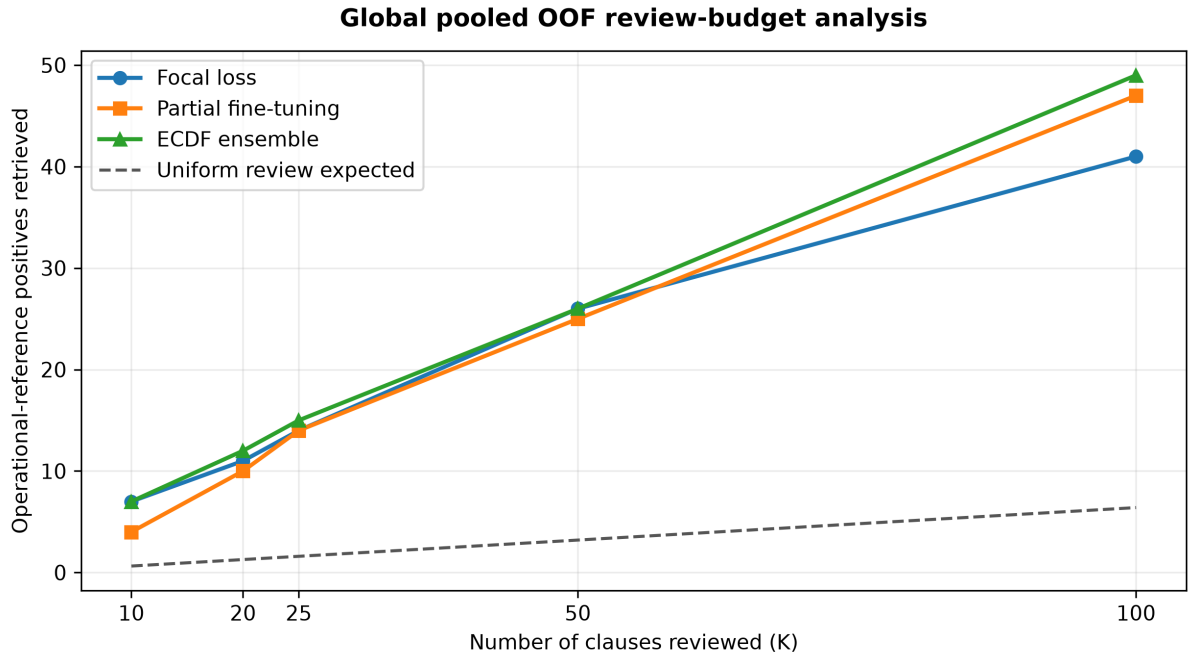


Figure 5.11: Reference-positive retrieval at pooled OOF review budgets for the earlier 2,125-row target-only transformer study. Counts are operational-reference based; the random line is the expected count under uniform sampling.

Table 5.12: Updated ensemble OOF AP by RFC family. Small-family estimates are especially uncertain.

Family	Rows	Positives	AP
HTTP/2bis	239	11	0.577
HTTP Semantics	385	34	0.796
QUIC HTTP	314	12	0.579
QUIC Recovery	162	11	0.731
QUIC TLS	189	8	0.571
QUIC Transport	794	57	0.646
TLS 1.3	54	3	0.917
Cache-Status	5	2	1.000
IPv6	1	0	–

Chapter 6

Discussion

6.1 Answers to the Proposal Research Questions

6.1.1 RQ1: Draft-Stage Detection Effectiveness

The experiments show that contemporaneous draft text contains useful ranking signal against the project's operational reference. On the latest matched classical cohort, engineered features plus TF–IDF reach AP 0.3584 with a Linear SVM at 6.60% prevalence. On the separate nine-family reviewed-label transformer cohort, the DeBERTa rank ensemble reaches AP 0.6736 at 6.44% prevalence and retrieves 19 of 20 reference positives in its pooled top-20 ranking. These estimates use only information available at the evaluated draft revision and isolate draft families across folds. The answer is nevertheless qualified: the cohorts are candidate-enriched, labels are not independent expert gold, the updated cohort contains 100 error-selected confirmations, family intervals overlap, and the latest classical rerun has already consumed the proposed TCP/DNS holdouts. RQ1 is therefore answered as a development-stage feasibility result, not as validated prospective accuracy.

6.1.2 RQ2: Revision History, Cross-References, and Calibration

The controlled classical ablation does not support a consistent revision-feature improvement. Revision features change AP by -0.0081, -0.0012, and -0.0015 for logistic regression, SVM, and random forest; corresponding F_2 changes are 0.0000, $+0.0110$, and $+0.0227$. Earlier predecessor-only models also remained near or below their cohort prevalence. Cross-reference counts are included among engineered features, but no isolated cross-reference ablation was completed. The revision experiment records no Brier score, expected calibration error, or calibration curve. Thus RQ2 is partly answered: the tested contemporaneous revision features do not improve ranking AP in the latest rerun, while

their thresholded F_2 effect is model dependent and the cross-reference-specific and calibration parts remain unresolved. Hypothesis H1 is not supported as a general claim.

6.1.3 RQ3: Comparison of Model Families

Table 6.1 summarizes the evidence without presenting incompatible cohorts as a common leaderboard. Fine-tuned DeBERTa provides the strongest observed ranking result in its own updated cohort. Classical engineered-plus-TF–IDF models provide competitive, inexpensive, and auditable rankings in their cohort. Direct Qwen prompting provides high recall but low precision, and its fitted lexical–semantic hybrid improves ranking within the Qwen study. The heuristic is most transparent but substantially weaker. Reproducibility is strongest for deterministic rules and classical pipelines and weaker for one-seed transformer and hardware-dependent local-LLM runs. Review-budget and referral analyses are available, but a common latency, energy, token-cost, or monetary-cost benchmark was not recorded. RQ3 is therefore answered for effectiveness and workload only under separate protocols; the quantitative latency/cost comparison remains incomplete.

Table 6.1: Proposal-RQ3 comparison. Scores are the strongest representative development result within each row's own cohort and must not be ranked across rows as if they shared labels and folds.

Approach	Representative evidence	Reproducibility	Efficiency/workload evidence
Fixed heuristic	AP 0.0873 on 2,425-row cohort	Deterministic rules and threshold	Negligible inference cost; weak ranking
Classical ML	SVM AP 0.3584; RF AP 0.3363; logistic AP 0.3265 on matched cohort	Fixed grouped folds and fold-local preprocessing	CPU-friendly; logistic recall 0.706 at precision 0.190
Fine-tuned transformer	Updated ECDF ensemble AP 0.6736 on 2,143 rows	Hash-pinned, but one training seed and overlapping family intervals	Top-20 retrieves 19 reference positives; no standardized latency/cost record
LLM-only	Qwen 8-shot AP 0.1960 on 2,142 rows	Frozen local model and cached outputs; hardware-sensitive	Recall 0.778 at precision 0.098 under raw rule; token/latency benchmark absent
Fitted hybrid	Qwen/TF–IDF/MiniLM V4 AP 0.3252 on 2,142 rows	Nested fitted components, but adaptively developed	Cross-fitted F_1 precision/recall 0.395/0.389; all-row LLM scores already computed

6.1.4 RQ4: Hybrid Routing and LLM-Call Reduction

The implemented V3/V4 hybrids combine cached Qwen scores with fitted lexical and semantic components; they are score-level hybrids rather than the proposal's conditional router. Because Qwen was evaluated for the cohort before combination, these experiments do not demonstrate fewer LLM calls. The separate referral analysis estimates how many clauses a classifier would send to human review at target recall, but it does not measure conditional LLM invocation. The fixed three-model rank ensemble similarly uses already-computed scores. RQ4 is therefore not answered in the intended causal comparison, and hypothesis H2 remains untested. A valid experiment must freeze a routing policy, count avoided LLM invocations, and compare effectiveness, latency, and cost against an LLM-only run on the same untouched cohort.

6.1.5 RQ5: Evidence and Explanation Quality

The local application can request optional Qwen explanations for selected clauses, and the prompting formats can emit rationales or structured fields. However, the repository contains no completed expert study of evidence-span correctness, ambiguity-subtype accuracy, rationale–label agreement, hallucinated-reference rate, usefulness, or clarity. Explanation text is not used as a prediction feature and is presented as a hypothesis to reviewers. RQ5 and hypothesis H3 therefore remain unanswered. This negative finding is important: classifier effectiveness cannot be used as a proxy for explanation faithfulness.

6.2 Interpretation of the Development Results

The updated family-isolated OOF scores indicate that clause text contains a useful ranking signal relative to the operational labels. Among the three methods evaluated on this same updated cohort, the rank ensemble has the highest observed AP and top-budget retrieval, but its family-cluster interval overlaps those of focal loss and partial fine-tuning. This ordering is descriptive, not a significance result. A second run across multiple seeds and an independently sampled, expert-adjudicated evaluation set are needed before claiming a reliable winner.

The separate language-model study shows a different pattern: direct Qwen prompting gives high recall at low precision in V1, while the stricter V2 prompt substantially reduces recall and AP. Adding a nested lexical classifier in V3 and frozen semantic features in V4 improves AP within that study's development cohort. These are adaptive comparisons with an operational label reference; they neither establish that LLMs outperform DeBERTa nor support direct numeric comparison across the two cohorts.

Ranking is more defensible than treating the model as a binary gate. The selected inner-fold thresholds optimize a precision constraint on fitting data, but pooled OOF behavior and family-specific outcomes show that the constraint does not transfer uniformly. A user-facing system should present clauses for expert review, expose source revision and surrounding context, and avoid claims that a low score certifies correctness.

On the separate classical/heuristic cohort, engineered features combined with TF–IDF produced stronger AP than the fixed heuristic and the tested embeddings-only baselines. This supports the value of lexical and explicit clause cues for ranking within that cohort, but the high-recall logistic operating point has low precision and does not remove the need for human review. The earlier LOFO results offer additional historical evidence for sparse lexical features, but differ in annotation, cohort, and validation setup. Frozen embeddings and basic predecessor features did not improve the matched LOFO baseline in the recorded feature setup; this is not evidence that semantic or temporal context is universally unhelpful. These classical and Qwen/hybrid results were not evaluated on the updated 2,143-row transformer cohort; cross-study AP values must not be treated as a unified model leaderboard.

The twelve newly evaluated revision features produce small changes in the matched classical cohort: AP decreases slightly for all three classifiers, while F_2 is unchanged for logistic regression and increases for SVM and random forest. The fixed three-component rank ensemble reaches AP 0.4619 on the label-compatible 2,097-row intersection, but the classical predictions were trained using earlier labels while V4 and the combination were developed on the intersection. Its narrow AP difference over the classical component is descriptive, not evidence of complementary generalization or a fair head-to-head win.

Across the older DeBERTa ablations, soft-target training did not improve the primary fold-balanced selection statistic and local context reduced it. A lower soft-target referral estimate at a fixed recall objective is a potentially useful future routing hypothesis, not evidence that it improves detector quality. The updated reviewed-label transformer run changes the operational labels and effective cohort relative to the earlier target-only study. The 100 human-confirmed cases were selected using historical OOF errors, and the associated family-bootstrap intervals overlap; consequently, the observed ensemble gain is not independent confirmation and may reflect the error-enriched evaluation design.

Revision alignment provides useful lineage and leakage-control infrastructure, but its pilot has not been evaluated against adjudicated gold links and its outputs are not used as predictive features in the transformer study.

6.3 Deployment Interpretation

The repository includes a local RFC-review browser prototype that currently applies the saved engineered-feature/TF–IDF logistic model only to clauses classified as normative by explicit uppercase BCP 14 keywords after parser exclusions. The newer transformer ensemble is presented as a research candidate, not wired for live single-RFC scoring. Lowercase-modal and section-context candidates are omitted rather than manually confirmed; consequently, RFCs lacking explicit BCP 14 terms may receive no predictions. Optional local Qwen explanations are restricted to selected flagged results and do not affect predictions. This conservative routing may reduce recall and is not a comprehensive normative-clause detector. The prototype is not an evaluated deployment system: the active model's score is not a calibrated probability and its saved threshold is not deployment-validated; generated explanations remain hypotheses. Automatic rejection or publication certification is inappropriate. The top-50 results in Tables 5.2 and the earlier transformer analysis are retrospective development diagnostics, not estimates of production precision or guaranteed reviewer yield.

6.4 Reproducibility and Governance

The experiment records hashes for the authoritative label source, effective dataset, model snapshot, folds, configuration, and derived predictions. The earlier source/effective-dataset SHA-256 values begin `8e395e83087c` and `19364a405048`; the updated reviewed-label source/effective-dataset values begin `8ffa654c2c40` and `4e20a2571bfe`. Full hashes remain in the experiment manifests. The model snapshot is used offline, and the experiment manifest pins seed, tokenizer, and training configuration. The analysis-repair version validates and reuses the parent experiment's 40 prediction artifacts; it does not retrain models or constitute a second evaluation. Code and selected aggregate tables are available in the project repository [17]; private annotation exports, row-level outputs, model weights, and large run directories are intentionally not part of a public release. The source annotation files required to reproduce the model results are therefore not present in a fresh repository clone, and the current result tables alone are insufficient for full independent reproduction.

6.5 Threats to Validity

6.5.1 Construct validity

The binary reference operationalizes ambiguity through a particular rubric and annotation workflow. It does not exhaust RFCScope's seven subtypes, and AI-supported labels

can reproduce shared model biases. Qwen3 contributed to annotation support, making the Qwen agreement analysis non-independent of label construction. The manual-conflict process is not equivalent to blinded, independent, double expert annotation. These issues limit claims about the real-world construct of ambiguity.

Candidate selection is also enriched rather than population-random. Metrics estimate performance on the reviewed candidate cohort, not on every sentence or clause in all Internet-Drafts. AP and prevalence therefore must not be transported to the full IETF corpus.

The multi-model consensus is not a statistically independent annotation panel. Low pairwise kappa and the large proportion of two-to-one positive votes indicate material uncertainty, while the model-target sensitivity analysis shows that the same predictions can score very differently under alternative annotator targets. No experiment in this paper estimates human inter-annotator reliability against an adjudicated expert gold set.

There is also an unresolved provenance/count discrepancy in the candidate exports: the evaluated LOFO artifact contains five errata-derived rows, all negative and all TLS, whereas an earlier progress report cited 121 errata positives and 242 TLS records. The corpus-level count of 31 verified technical errata refers to source documents, not positive clause labels, and does not reconcile those candidate-export counts. Until the row-level source definition is reconciled, errata-derived candidate results must not be interpreted as a comprehensive errata evaluation.

6.5.2 Internal validity

Family splits, fitting-only preprocessing, inner-only thresholds, and exact/near/lineage exclusion reduce leakage. They cannot eliminate pretraining contamination: modern encoders may have seen historical RFCs or drafts. The experiment uses a single seed, and family bootstrapping does not account for training variance. Architecture and analysis choices were made within the wider development process; the reported OOF study should therefore be interpreted as development evidence, not a pre-registered confirmatory benchmark.

The updated 100-case human-confirmed annotation subset was selected from historical OOF errors. It is therefore subject to error-selection bias; evaluating models on these cases or their families cannot provide an unbiased estimate of generalization or improvement. The reported updated-cohort OOF metrics are complete, but are not independent confirmation: the analysis-repair step validates and reuses predictions from the parent training run. A future assessment requires an independently sampled, untouched evaluation set, preferably spanning families absent from model selection.

6.5.3 External validity

Only nine families contribute scored rows to the updated transformer development study, and QUIC/HTTP dominate row counts. The smallest families cannot independently support meaningful AP estimates. TCP and DNS are absent from that transformer study, but their labels were consumed by the latest all-family classical rerun; consequently, they are neither transformer evidence nor untouched project-level tests. Routing protocols with historical drafts outside the current cohort, standards from other organizations, and newly authored drafts after the models' pretraining period are also untested.

6.5.4 Alignment validity

The TLS alignment run has deterministic audits but no reviewed gold links. Alignment confidence is rule-based and uncalibrated. Split/merge grouping, move heuristics, and long-clause pooling are approximations. Accordingly, correspondence statistics are engineering coverage, not evidence of alignment accuracy, and automatic revision relationships are not used as ground-truth ambiguity labels.

6.6 Ethical and Practical Considerations

All source standards and errata are public technical documents. The principal risk is epistemic rather than personal: an incorrect "ambiguous" flag could waste reviewer time or unfairly characterize an author's work, while a false negative could create unwarranted confidence. The system must expose uncertainty, provenance, and its non-certification role. Generated explanations should be treated as hypotheses and checked against the exact contemporaneous text. Final evaluation should be reported once, including negative outcomes, and the sealed set should not be recycled for iterative tuning.

Chapter 7

Conclusion and Future Work

7.1 Summary of Achievements

DraftAmbiguity combines corpus acquisition, structural parsing, normative-clause segmentation, revision alignment, lineage-aware candidate construction, AI-assisted consensus annotation, and grouped classical/transformer/LLM evaluation. The project reports correspondence coverage separately from ambiguity classification and excludes later revisions, RFC endpoints, errata, annotation votes, and rationales from model features. Results span several annotation snapshots and protocols. In the latest updated-label transformer study, focal loss, partial fine-tuning, and their equal-weight fitting-side ECDF ensemble were evaluated on the same 2,143-row, nine-family cohort (138 positives): AP was 0.5823, 0.5761, and 0.6736, respectively. The family-cluster intervals overlap, only one training seed was run, and 100 human-confirmed records were selected from historical OOF errors; the observed ensemble lead is therefore descriptive and potentially affected by error-selection bias. Other studies reported AP 0.0603 for weighted DeBERTa on an initial low-prevalence cohort; 0.291 for TF–IDF-plus-engineered features in an earlier nine-family LOFO study; 0.3584 for the latest all-family classical SVM; 0.3715 for a nested DeBERTa ensemble; and 0.1960 for eight-shot Qwen and 0.3252 for its V4 lexical/semantic hybrid. A fixed three-model rank ensemble reached AP 0.4619 on a label-compatible intersection, but remains exploratory because its component predictions have mismatched label provenance. These cross-study values are not a unified leaderboard. The local review GUI currently runs the saved classical logistic-regression artifact; the newer transformer ensemble is a research candidate, and optional local Qwen explanations do not change predictions. The GUI and all model outputs remain unvalidated for independent deployment. The attempted second-generation annotation panel produced no corpus labels because it was blocked at preflight.

The strongest conclusions are methodological rather than deployment claims: family-isolated evaluation is feasible; target text and transparent lexical/engineered features

provide useful ranking signal against the operational reference; the tested revision features do not improve ranking AP in the latest matched rerun; and performance depends on model family, data snapshot, evaluation protocol, and annotation target. The low inter-model agreement, adaptive development comparisons, overlapping family uncertainty, single-seed transformer results, error-selected human review, absence of an independent expert gold set, and loss of the proposed TCP/DNS holdouts prevent a claim of validated ambiguity-detection accuracy. Annotation sources and large run artifacts are also not included in a fresh clone.

7.2 Future Work

Before making a deployment or final-generalization claim, the project should: (1) independently double-annotate and adjudicate a stratified expert sample; (2) repeat the selected transformer configurations across multiple seeds; (3) implement the proposal's actual conditional LLM router and record calls, latency, energy or token use, and monetary cost against a same-cohort LLM-only baseline; (4) isolate cross-reference features and add Brier score, expected calibration error, and reliability diagrams; (5) conduct blinded expert assessment of evidence spans, subtype assignments, rationale faithfulness, hallucinated references, usefulness, and clarity; (6) validate revision links, including split/merge relationships, against a reviewed alignment sample; and (7) preregister and preserve a new genuinely untouched evaluation, such as unseen draft families or later time-sliced revisions, because TCP and DNS have already entered the classical evaluation. Until those steps are complete, the appropriate claim is a reproducible, promising development-stage review aid—not a published-standard oracle or autonomous quality gate.

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Appendix A

Research-Question and Hypothesis Traceability

Table A.1 records the disposition of every research question in the approved proposal. "Partly answered" means that valid repository evidence covers some requested outcomes but not the complete question.

Table A.1: Traceability from proposal research questions to report evidence.

ID	Proposal question	Status	Evidence and conclusion
RQ1	Draft-stage effectiveness using only contemporaneous information	Partly answered	Grouped classical and transformer studies show useful ranking signal; candidate enrichment, error-selected review, and consumption of the planned TCP/DNS holdouts prevent a prospective accuracy claim.
RQ2	Revision/cross-reference effects on F_2, PR-AUC, and calibration	Partly answered	Matched revision ablation has small mixed AP/F_2 effects. Cross-reference isolation and calibration metrics were not completed.

ID	Proposal question	Status	Evidence and conclusion
RQ3	Effectiveness, reproducibility, latency, cost, and workload across method families	Partly answered	Effectiveness and review workload are reported under separated cohorts. Reproducibility limitations are documented. No standardized latency/cost benchmark exists.
RQ4	Hybrid effectiveness with fewer LLM calls	Not answered as proposed	Implemented hybrids combine scores computed for all rows; conditional-call savings were not measured.
RQ5	Faithfulness/usefulness of evidence spans, subtypes, and explanations	Not answered	Optional explanations exist, but no expert explanation-quality evaluation was completed.

Table A.2: Disposition of proposal hypotheses.

ID	Hypothesis	Outcome	Reason
H1	Revision history improves F_2 and PR-AUC	Not supported generally	Changes are small and mixed across the three classical models.
H2	Hybrid routing retains effectiveness while reducing LLM calls	Not tested	Score-level hybrids do not implement conditional LLM routing.
H3	Evidence grounding reduces unsupported explanations	Not tested	No controlled explanation-faithfulness assessment was completed.
H4	Random splitting is more optimistic than family/temporal splitting	Not directly tested	The main studies correctly use family isolation, but no matched random-versus-grouped experiment quantifies the difference.

Appendix B

Reproduction and User Instructions

B.1 Revision-Augmented Classical Experiments

Run the following commands from the repository root. They reproduce the current revision-augmented classical runs and comparison plots when the tracked annotation and matcher-cache inputs are present.

```
python -m models.classical.train_annotation_engineered_tfidf_revision
python -m models.classical.annotation_classifier_experiments \
  --classifier linear_svm --feature-mode engineered_tfidf_revision
python -m models.classical.annotation_classifier_experiments \
  --classifier random_forest --feature-mode engineered_tfidf_revision
python -m models.classical.plot_annotation_results
python paper/generate_report_figures.py
```

The principal outputs are stored in:

- `results/ML/v2_engineered_tfidf_revision/`;
- `results/ML/additional_classifiers/`; and
- `results/ML/figures/`.

Each revision run includes OOF predictions, fold metrics, raw revision features, a final development model, and a provenance manifest.

B.2 Local Review Application

After restoring the approved classical model artifact described in the repository README, start the local application with:

```
python -m gui.app
```

The interface binds to `http://127.0.0.1:8765/`. A reviewer can upload a rendered RFC text file, inspect positive-ranked clauses, filter by section, download results, and optionally request local explanations for at most five selected clauses. Scores are not calibrated probabilities and results require human verification.

B.3 Report Build and Validation

Generate the report-specific figures first, then compile the bibliography and report from the `paper/` directory:

```
python paper/generate_report_figures.py
cd paper
pdflatex main
bibtex main
pdflatex main
pdflatex main
```

Focused regression tests for the revision features, classical runners, and result plotting are:

```
python -m pytest -q tests/test_revision_features.py \
  tests/test_annotation_logreg_experiment2.py \
  tests/test_annotation_classifier_experiments.py \
  tests/test_plot_annotation_results.py
```